6221 tutor

yyyy

quiz

Last Year:

7 true/false questions (10 marks per question)

1 multiple choice question (30 marks)

Total: 100 marks. 15% of your final score

Please note that all questions require a brief explanation of the reasoning.

For reference only

1. Can 8 points be used to obtain an essential matrix?
2. Can 4 coplanar points be used to obtain a Homography matrix?
3. Can both Essential matrix (E) and Homography matrix (H) be used to obtain R and a scaled translation x?
4. Precision describes the difference between the measurement and true value, while accuracy relates to reproducibility of sensor results.
5. Is Kalman filtering statistically optimal for linear systems?
6. If two random variables, X and Y, are (uncorrelated) independent, the variance of their sum equal to the sum of their individual variances.
7. Structure from Motion (SFM) is the study of the rotation and translation that separates these two camera views and the relative structure of the observed feature points

Question answer

1. Can 8 points be used to obtain an essential matrix?
2. Can 4 coplanar points be used to obtain a Homography matrix?
3. Can both Essential matrix (E) and Homography matrix (H) be used to obtain R and a scaled translation x?

We begin with the Essential Matrix and 8 Point Algorithm. Assume at least 8 points are matched in two images, no four 3D points are coplanar.

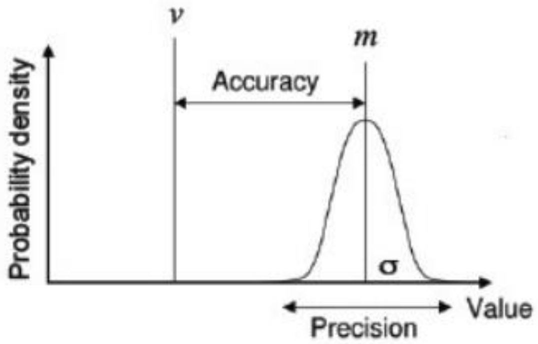
We now address the issue of planar scenes using the Euclidean Homography Matrix and the 4 point algorithm Assume at least 4 coplanar, noncollinear points are matched in two images

	Essential Matrix	Homography Matrix
points	8	4
requirements	no coplanar	coplanar, non-collinear
translation	nonzero translation	NA
outcomes	unique	depend on experience
accuracy	NA	more accurate

Question answer

4. Precision describes the difference between the measurement and true value, while Accuracy relates to reproducibility of sensor results

- Accuracy (exactitude, opposite of error)
  - degree of conformity between the measurement and true value
  - often expressed as a portion of the true value: *Accuracy=1-|m-v|/v, m = measured value, v = true value*
- Precision (or repeatability)
  - relates to reproducibility of sensor results: precision=range divided by standard deviation



Lec1， page24

Question answer

5. Is Kalman filtering statistically optimal for linear systems?

optimal in a statistical sense. *If the system can be described with a linear model and both the system and sensor error can be modeled as white Gaussian noise, the Kalman filter will provide* ***unique statistically optimal estimates*** *for the fused data.*

Lec4， page17

The Kalman filter is optimal if the following assumptions are true:

1) The system is linear;
2) The process and measurement noise are uni-modal Gaussian.

Lec4， page45

Question answer

6. If two random variables, X and Y, are (uncorrelated) independent, the variance of their sum equal to the sum of their individual variances.

$V(X + Y) = V(X) + V(Y)$ for X and Y uncorrelated

Proof :

from page 12

$V(X + Y)$

$= E[(X + Y)^2] - (E[X + Y])^2$

$= E[(X^2 + Y^2 + 2XY] - (E[X] + E[Y])^2$

$= E[X^2] + E[Y^2] + 2E[XY] - (E[X])^2 - (E[Y])^2 - 2E[X]E[Y]$

$= E[X^2] + E[Y^2] - (E[X])^2 - (E[Y])^2 + 2E[XY] - 2E[X]E[Y]$

$= E[X^2] - (E[X])^2 + E[(Y)^2] - (E[Y])^2 + 2Cov(XY)$

$= V(X) + V(Y)$

since $Cov(XY) = 0$ if X and Y are uncorrelated.

Lec4， page15

Question answer

7. Structure from Motion is the study of the rotation and translation that separates these two camera views and the relative structure of the observed feature points.

Structure From Motion - Introduction

- Given two images of matched points, it is possible to determine:
  - The rotation and translation that separates these two camera views
  - The relative structure of the feature points being viewed

Lec3， page4

Multiple choice question

Which of the following are advantages of multi-sensor integration and fusion? Justify your answer in detail.

- Redundancy
- Accuracy
- Reliability

Lec1 Page12-15

The Role of Multi-Sensor Integration and Fusion

The purpose of external sensors is to provide a system with useful information concerning some features of interest in the system's environment. This enables the machine to ***act intelligently*** via some intelligent programme.

The potential advantages in ***integrating*** and/or ***fusing information*** from ***multiple sensors*** are that the information can be decided more accurately, concerning features that are impossible to perceive with individual sensors, as well as in less time, and at a lesser cost.

The Role of Multi-Sensor Integration and Fusion

Complementary information from multiple sensors allows features in the environment to be perceived that are impossible to perceive using just the information from each individual sensor operating separately.

More timely information, as compared to the speed at which it could be provided by a single sensor, may be provided by multiple sensors either because of the actual speed of operation of each sensor, or because of the ***processing parallelism*** that may possibly be achieved as part of tile integration process.

The Role of Multi-Sensor Integration and Fusion

Redundant information is provided from a group of sensors (or a single sensor over time) when each sensor is perceiving, possibly with a different fidelity, the same features in the environment.

The integration or fusion of redundant information ***can reduce overall uncertainty*** and thus serve to increase the accuracy with which the features are perceived by the system.

Multiple sensors providing redundant information can also serve to ***increase reliability*** in the case of sensor error or failure.

The Role of Multi-Sensor Integration and Fusion

Less costly information, in the context of a system with multiple sensors, is information obtained at a lesser cost as compared to the equivalent quality of information that could be obtained from a single sensor.

Unless the information provided by the single sensor is being used for additional functions in the system, the total cost of the single sensor should be compared to the total cost of the integrated multi-sensor system.

With redundancy, it is possible then to ***use lower cost sensors with lower quality*** that can be combined to achieve the same quality of reading from a high-quality sensor (thereby being costly).

3 Mobile Robots

3. (a) A mobile platform is shown in Figure 2 The platform has one steered standard wheel and two standard wheels. A local reference frame (x_r, y_r) is assigned as shown in the figure. The radius of each standard wheel is 10 cm. If the rotational velocities of the steered standard wheel and the two standard wheels are denoted by $\dot{\varphi}_{ss}$, $\dot{\varphi}_{s1}$, $\dot{\varphi}_{s2}$ respectively, derive the rolling and sliding constraints of the mobile platform.

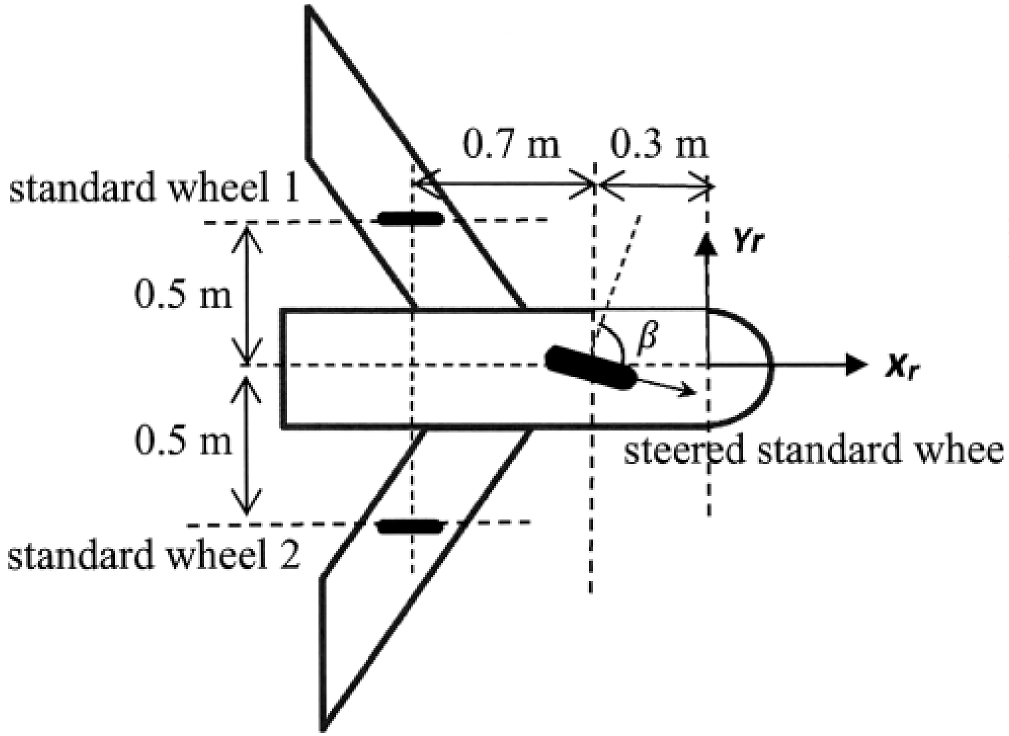
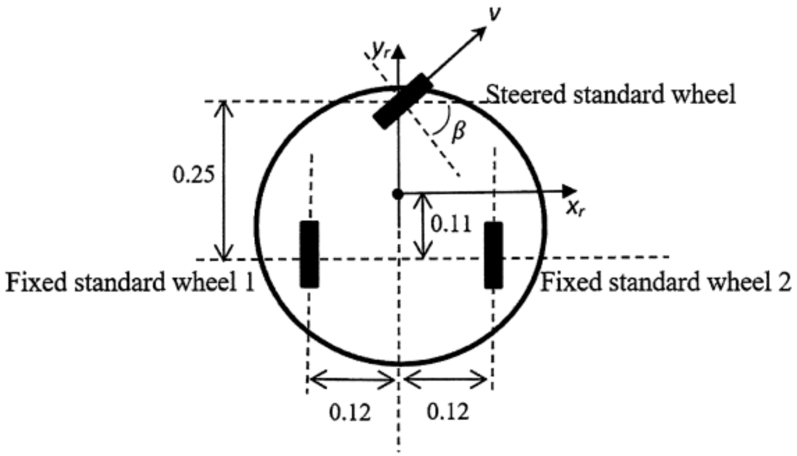


Figure 2

A mobile robot with two standard wheels and one steered standard wheel is shown in Figure 3. A local reference frame (x_r, y_r) and a steered angle β are assigned to the mobile robot as shown in Figure 3. The radius of each standard wheel is 5.0 cm and the radius of the steered standard wheel is 5.5 cm. If the rotational velocities of the steered standard wheel and the two standard wheels are denoted by $\dot{\varphi}_{ss}$, $\dot{\varphi}_{fs1}$, and $\dot{\varphi}_{fs2}$, respectively, derive the rolling and sliding constraints of the mobile robot.

(10 Marks)



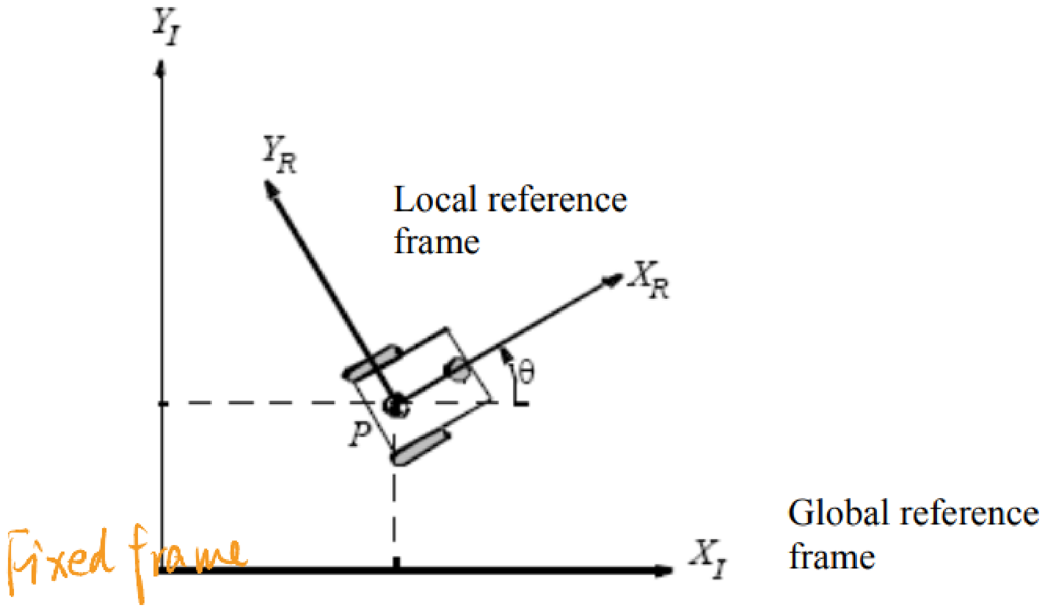
Note: all lengths are in meters.

Figure 3

Mobile Robots

To specify the position of the robot on the plane, a relationship between the **global reference frame** of the plane and the **local reference frame** of the robot is established.

Mobile frame



Global reference frame: $\{X_I, Y_I\}$

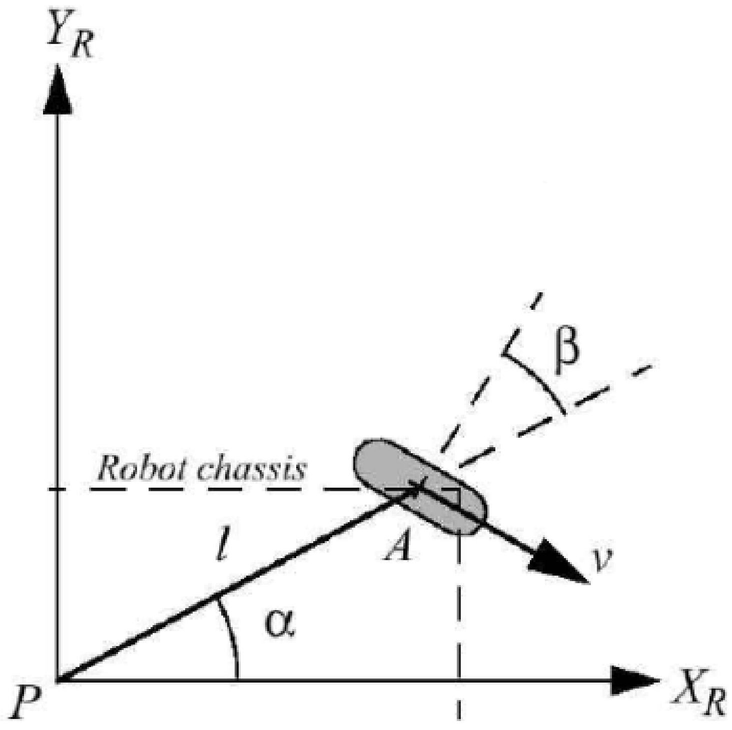
Local reference frame: $\{X_R, Y_R\}$

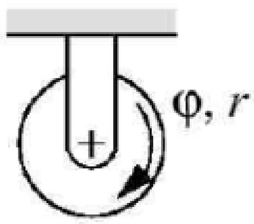
Basic wheel types

- **Standard wheel**
- **Castor wheel**
- Swedish wheel
- Ball or spherical wheel

Fixed standard wheel

Consider a fixed standard wheel *A* and indicates its position pose relative to the robot's local reference frame $\{X_R, Y_R\}$.





$\dot{\varphi} r = v$

The position of ***A*** is expressed in polar coordinates by *distance **l*** and *angle **α***.

The *angle of the wheel plane relative to the chassis* is denoted by $\boldsymbol{\beta}$, which is fixed.

The wheel, which has radius ***r***, and its rotational position is a function of time ***t***: $\boldsymbol{\varphi(t)}$.

Fixed standard wheel

The **rolling constraint** for this wheel is,

$$v = \dot{\varphi}\ r = \dot{x}_r \sin(\alpha+\beta) - \dot{y}_r \cos(\alpha+\beta) - \dot{\theta}\ l \cos\beta$$

$$= \begin{bmatrix} \sin(\alpha+\beta) & -\cos(\alpha+\beta) & -l\cos\beta \end{bmatrix} \boxed{\begin{bmatrix} \dot{x}_r \\ \dot{y}_r \\ \dot{\theta} \end{bmatrix}}$$

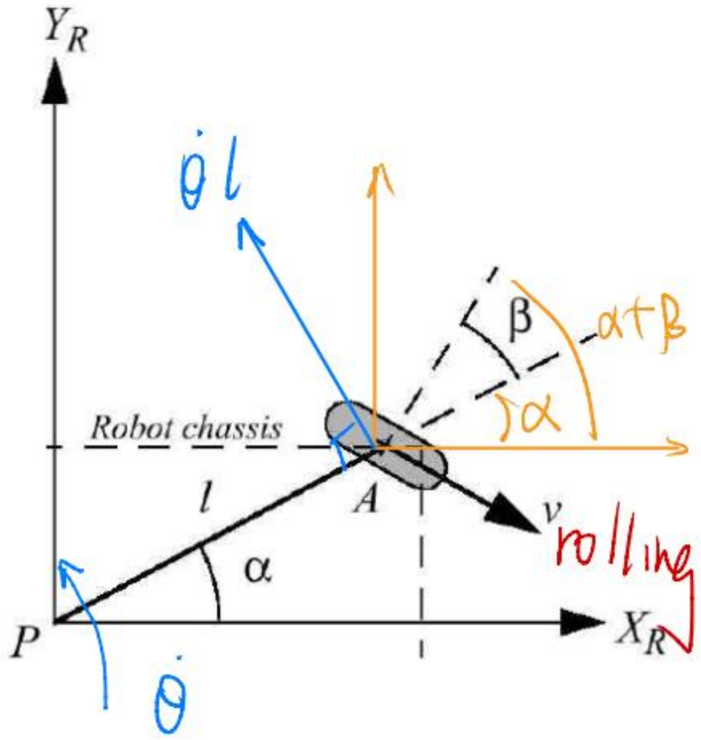
The above *rolling constraint* can be written as:

$$\begin{bmatrix} \sin(\alpha+\beta) & -\cos(\alpha+\beta) & -l\cos\beta \end{bmatrix} \boxed{R(\theta)\dot{\xi}_I} - r\dot{\varphi} = 0$$

or

$$j\ R(\theta)\dot{\xi}_I - r\dot{\varphi} = 0$$

where $j = \begin{bmatrix} \sin(\alpha+\beta) & -\cos(\alpha+\beta) & -l\cos\beta \end{bmatrix}$.



Fixed standard wheel

The **sliding constraint** for this wheel is,

$$\dot{x}_r \cos(\alpha+\beta)+\dot{y}_r \sin(\alpha+\beta)+\dot{\theta}\ l \sin\beta=0$$

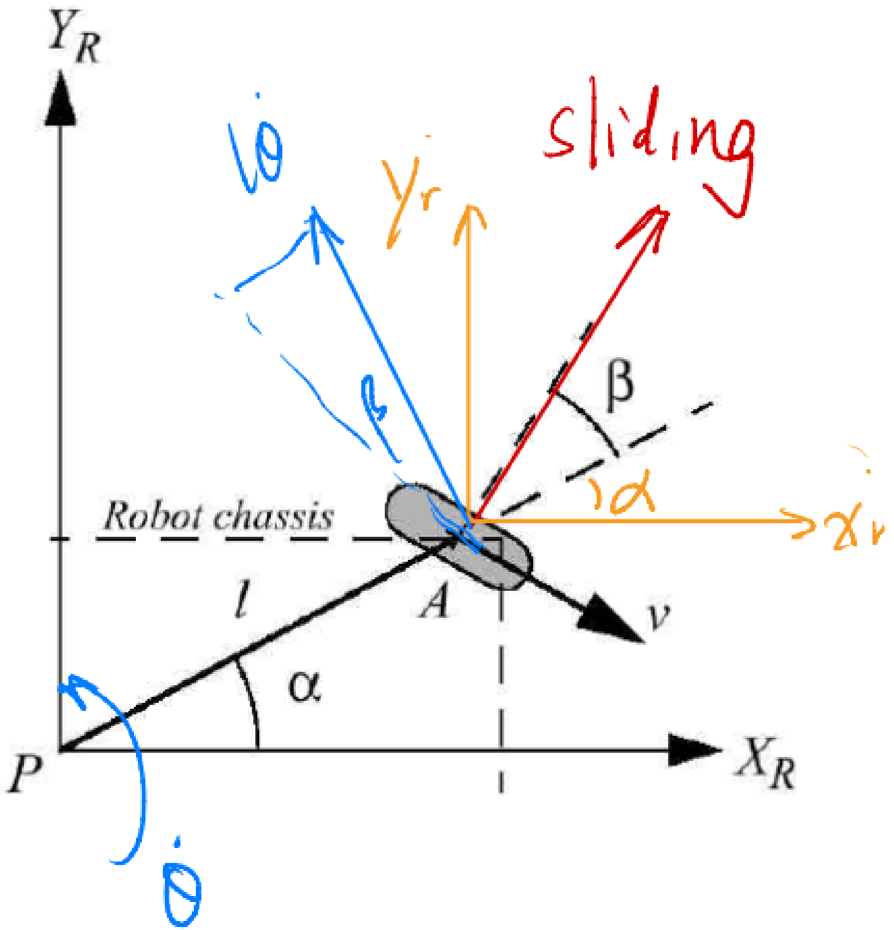
The above *sliding constraint* can be written as:

$$[\cos(\alpha+\beta)\quad \sin(\alpha+\beta)\quad l\sin\beta]R(\theta)\dot{\xi}_I=0$$

or

$$c\ R(\theta)\dot{\xi}_I=0$$

where $c=[\cos(\alpha+\beta)\quad \sin(\alpha+\beta)\quad l\sin\beta]$.



Fixed standard wheel

The above *rolling constraint* can be written as:

$$[\sin(\alpha+\beta) \quad -\cos(\alpha+\beta) \quad -l\cos\beta]R(\theta)\dot{\xi}_I - r\dot{\varphi} = 0$$

The above *sliding constraint* can be written as:

$$[\cos(\alpha+\beta) \quad \sin(\alpha+\beta) \quad l\sin\beta]R(\theta)\dot{\xi}_I = 0$$

Example 2: ① find α, β

Derive the rolling and sliding constraints for wheel 1.

Page 23

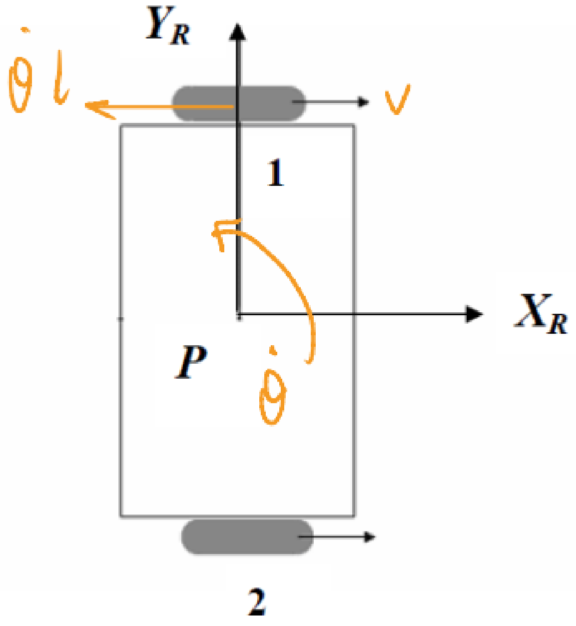
Note that wheel 1 is in a position such that $\alpha = \frac{\pi}{2}$, $\beta = 0$.

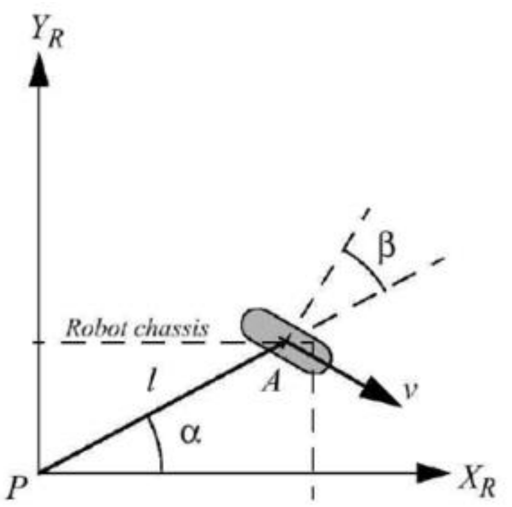
The rolling constraint for wheel 1 reduces to

$$\dot{\phi}\, r = \dot{x}_r \sin(\frac{\pi}{2}) - \dot{y}_r \cos(\frac{\pi}{2}) - \dot{\theta}\, l\cos 0 \Rightarrow [1 \quad 0 \quad -l]\begin{bmatrix}\dot{x}_R \\ \dot{y}_R \\ \dot{\theta}\end{bmatrix} - r\dot{\phi} = 0$$

The sliding constraint for wheel *1* is

$$\dot{x}_r \cos(\frac{\pi}{2}) + \dot{y}_r \sin(\frac{\pi}{2}) + \dot{\theta}\, l\sin 0 = 0 \Rightarrow [0 \quad 1 \quad 0]\begin{bmatrix}\dot{x}_R \\ \dot{y}_R \\ \dot{\theta}\end{bmatrix} = 0$$



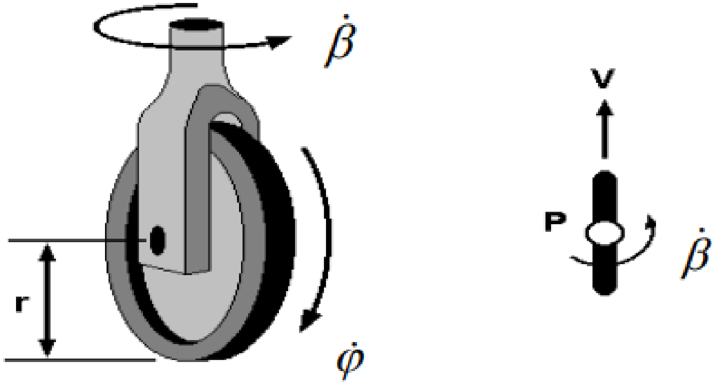


$\dot{\varphi} r = \dot{x}_R - \dot{\theta} l$ $\quad$ $\dot{y}_R = 0$

Steered standard wheel

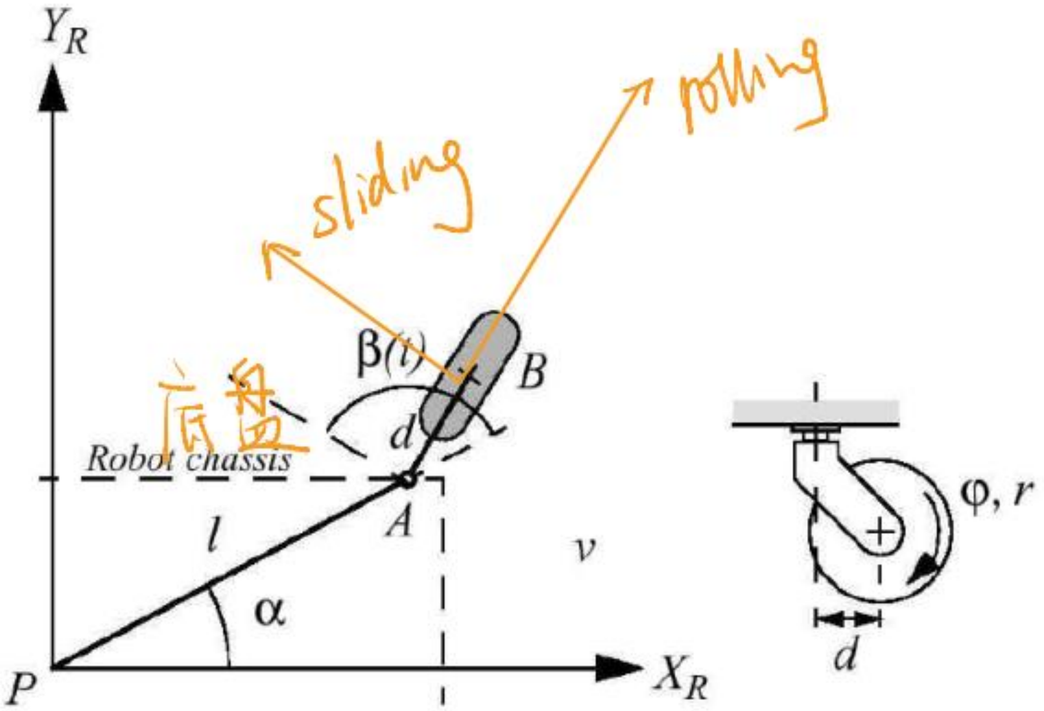
Steered standard wheel

$\dot{\beta}$ can change



The steered standard wheel differs from the fixed standard wheel only in that the wheel may rotate around a vertical axis passing through the center of the wheel and the ground contact point.

Castor wheel



The wheel *contact point is now at position B*, which is connected by a rigid rod *AB* of fixed length *d* to point *A*.

Castor wheel

For the castor wheel, the rolling constraint is identical to the standard wheel because the offset axis plays no role during motion that is *aligned with the wheel plane*:

$$[sin(\alpha+\beta) \quad -cos(\alpha+\beta) \quad -l\cos\beta]R(\theta)\dot{\xi}_I - r\dot{\varphi} = 0$$

or

$$j(\beta)R(\theta)\dot{\xi}_I - r\dot{\varphi} = 0$$

The castor geometry does, however, have significant impact on the sliding constraint.

$$[cos(\alpha+\beta) \quad sin(\alpha+\beta) \quad l\sin\beta]R(\theta)\dot{\xi}_I + d\dot{\beta} = 0$$

or

$$c(\beta)R(\theta)\dot{\xi}_I + d\dot{\beta} = 0$$

Formula

$$J(\beta) = [\sin(\alpha+\beta), -\cos(\alpha+\beta), -l\cos(\beta)]$$
$$C(\beta) = [\cos(\alpha+\beta), \sin(\alpha+\beta), l\sin(\beta)]$$

Standard Wheel & Steered Standard Wheel:

Rolling Constraint:

$$J(\beta)R(\theta)\dot{\xi}_I - r_{ss}\dot{\phi}_{ss} = 0$$

Sliding Constraint:

$$C(\beta)R(\theta)\dot{\xi}_I = 0$$

Caster Wheel:

Rolling Constraint:

$$J(\beta)R(\theta)\dot{\xi}_I - r_{ss}\dot{\phi}_{ss} = 0$$

Sliding Constraint:

$$C(\beta)R(\theta)\dot{\xi}_I + d\dot{\beta} = 0$$

Parameters:

$\alpha + \beta$

β

ℓ

r

ϕ

d

24-25 S1 Q3

3. (a) A mobile platform is shown in Figure 2 The platform has one steered standard wheel and two standard wheels. A local reference frame (x_r, y_r) is assigned as shown in the figure. The radius of each standard wheel is 10 cm. If the rotational velocities of the steered standard wheel and the two standard wheels are denoted by $\dot{\varphi}_{ss}$, $\dot{\varphi}_{s1}$, $\dot{\varphi}_{s2}$ respectively, derive the rolling and sliding constraints of the mobile platform.

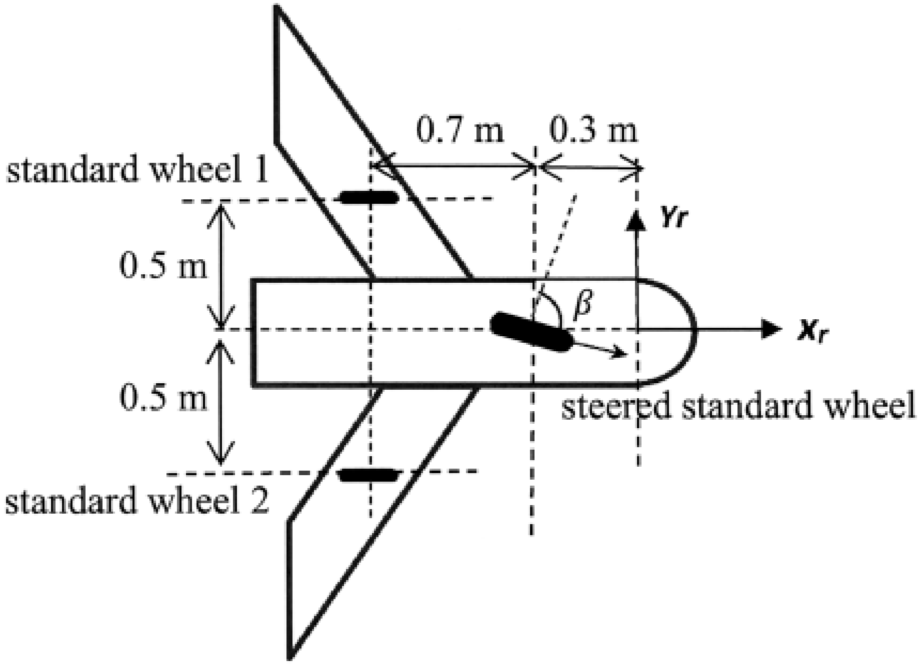


Figure 2

(8 Marks)

3. (a) A mobile platform is shown in Figure 2 The platform has one steered standard wheel and two standard wheels. A local reference frame (x_r, y_r) is assigned as shown in the figure. The radius of each standard wheel is 10 cm. If the rotational velocities of the steered standard wheel and the two standard wheels are denoted by $\dot{\varphi}_{SS}$, $\dot{\varphi}_{s1}$, $\dot{\varphi}_{s2}$ respectively, derive the rolling and sliding constraints of the mobile platform.

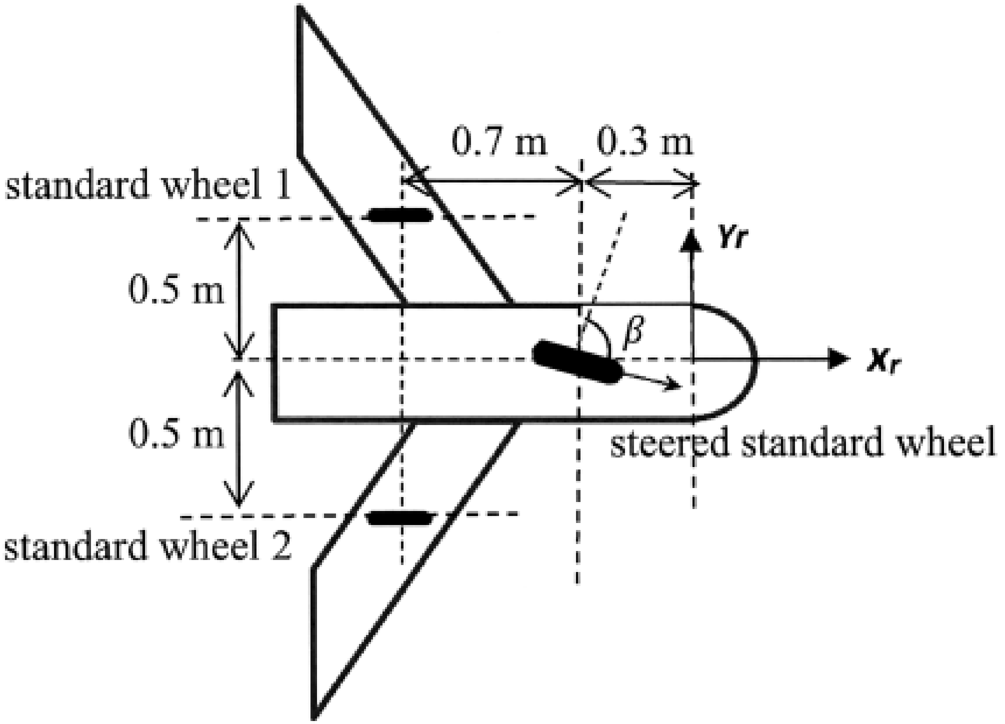


Figure 2

$$J(\beta) = [\sin(\alpha+\beta), -\cos(\alpha+\beta), -l\cos(\beta)]$$
$$C(\beta) = [\cos(\alpha+\beta), \sin(\alpha+\beta), l\sin(\beta)]$$

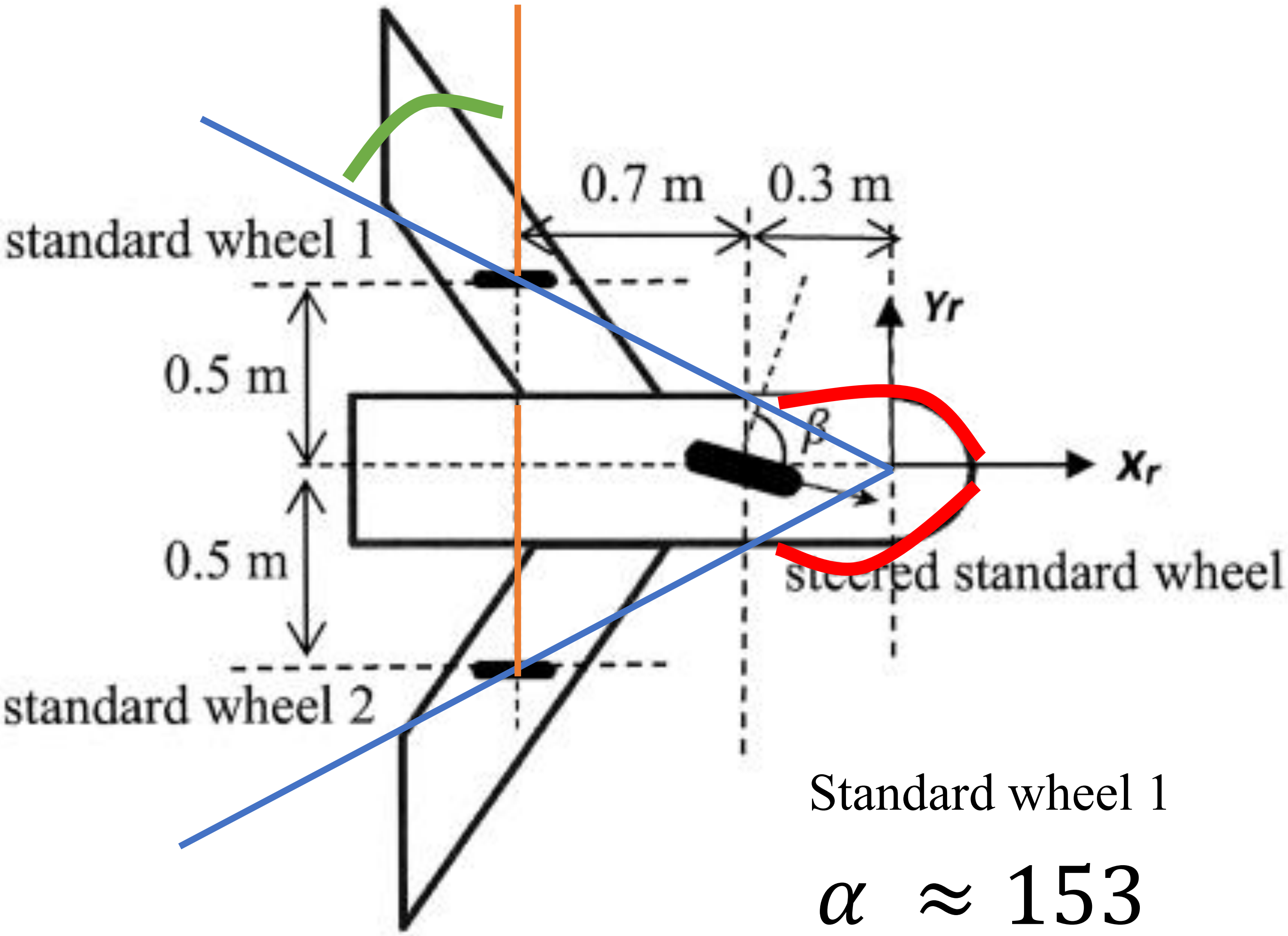
Standard Wheel & Steered Standard Wheel:

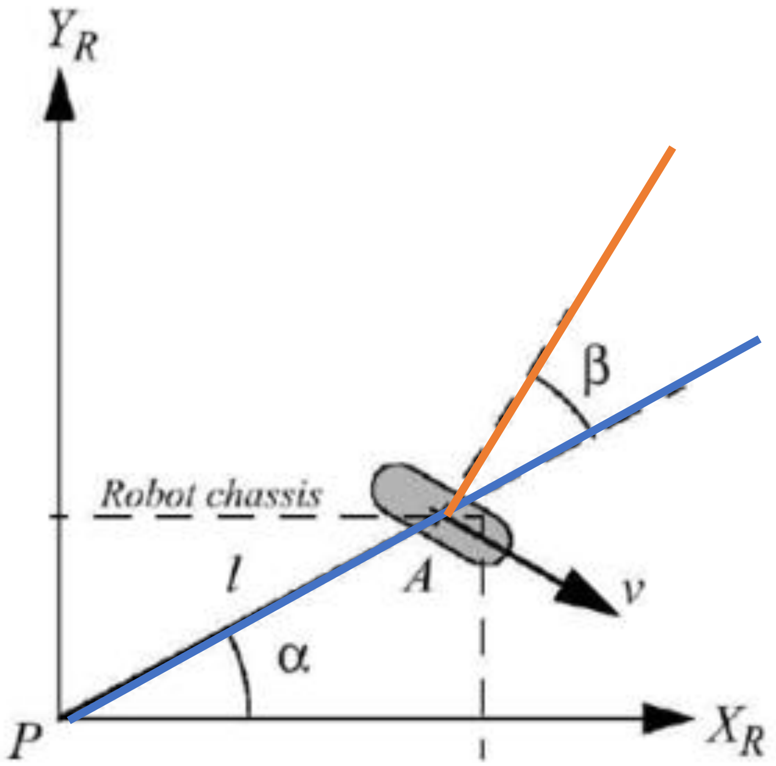
Rolling Constraint:

$$J(\beta)R(\theta)\dot{\xi}_I - r_{ss}\dot{\phi}_{ss} = 0$$

Sliding Constraint:

$$C(\beta)R(\theta)\dot{\xi}_I = 0$$

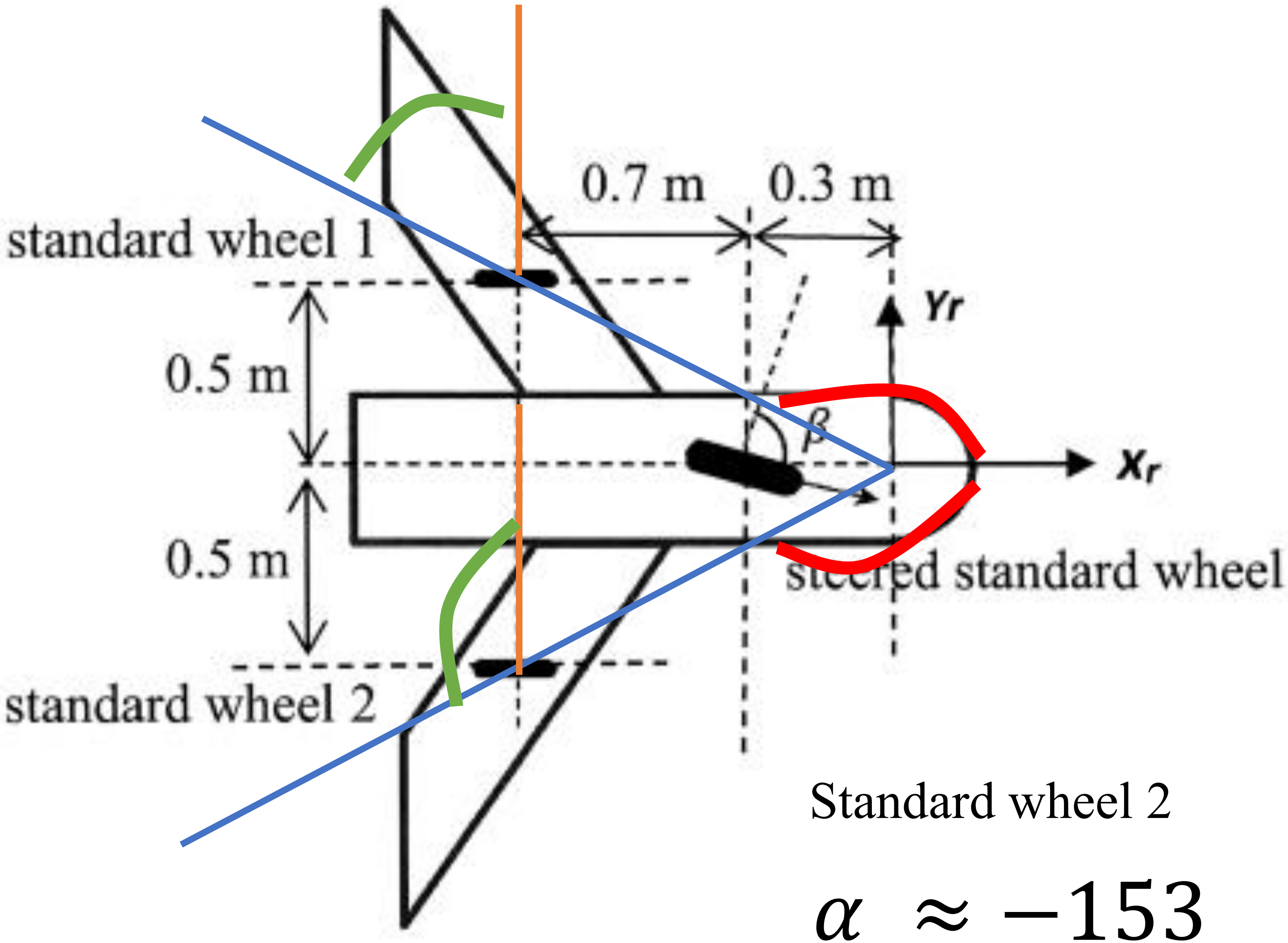


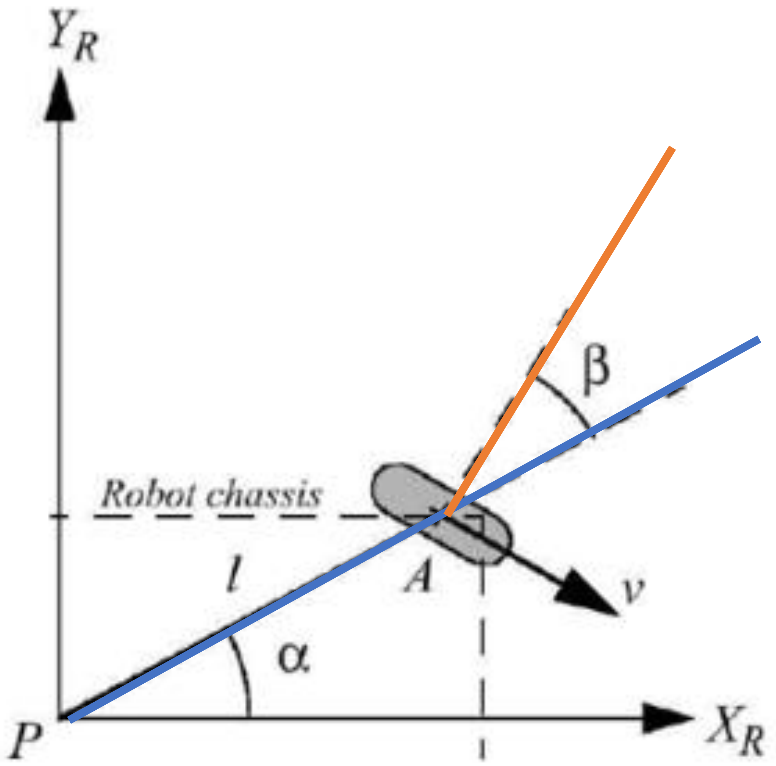


Standard wheel 1

$$\alpha \approx 153$$

$$\beta \approx -63$$

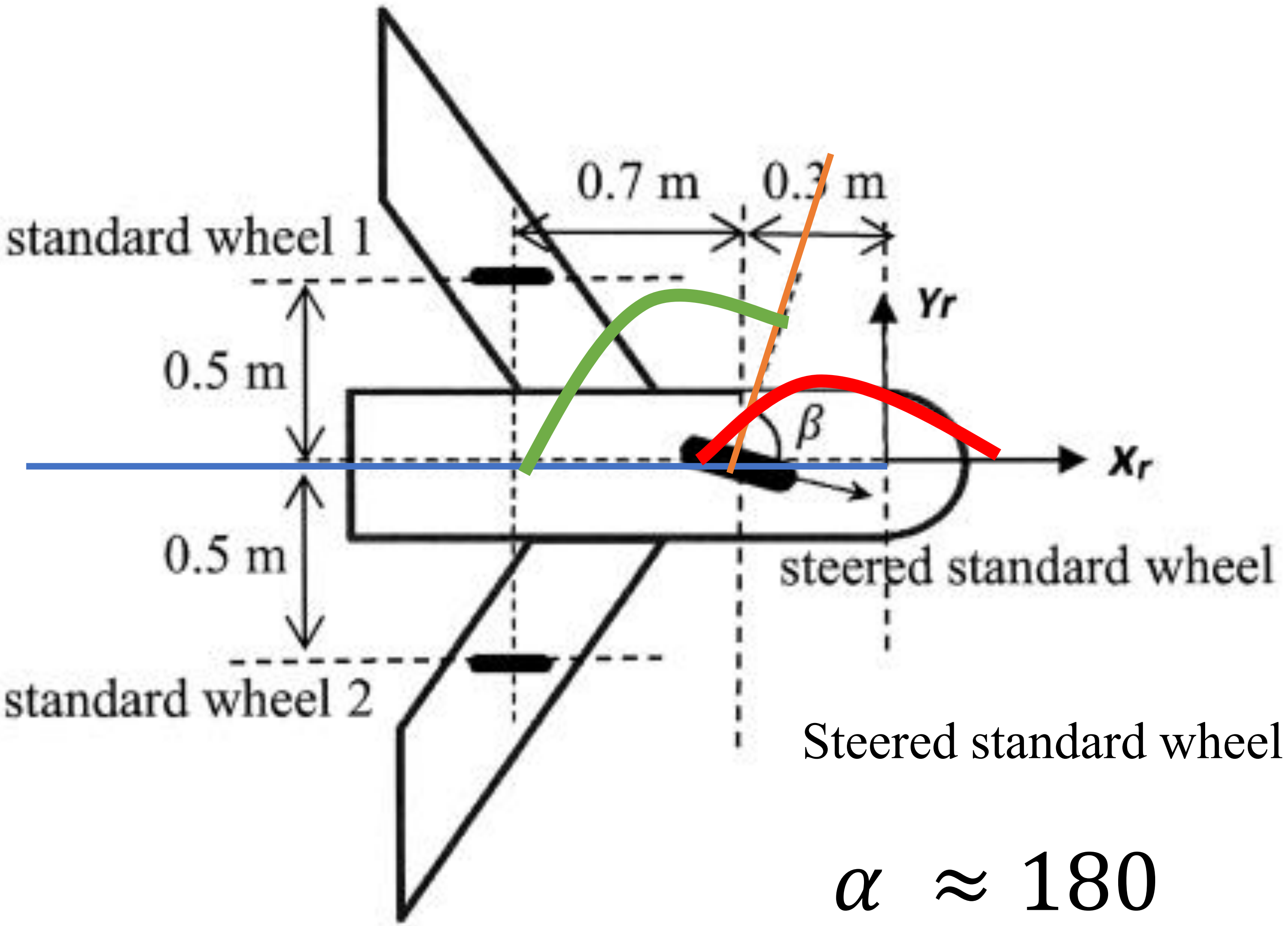


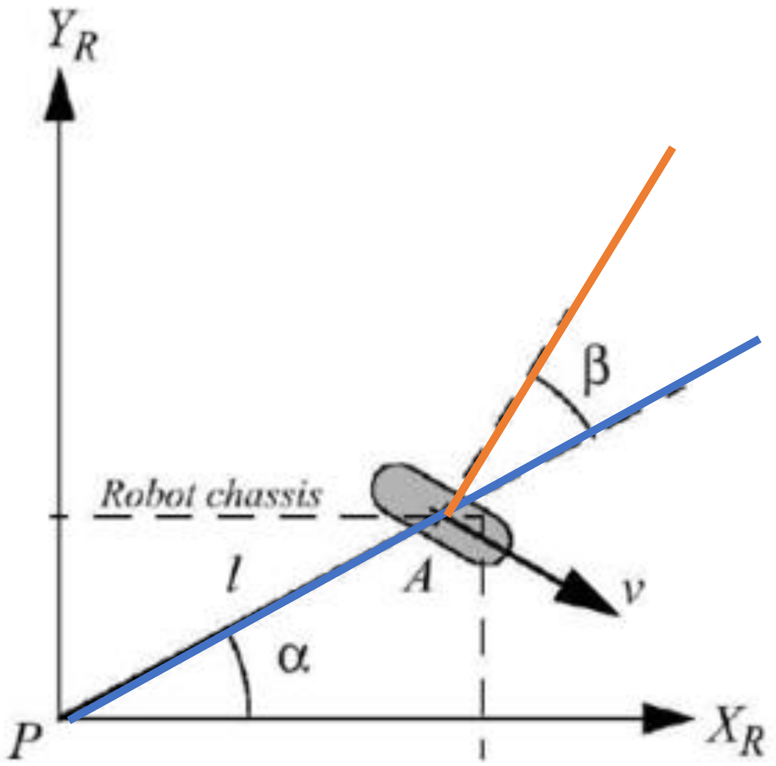


Standard wheel 2

$$\alpha \approx -153$$

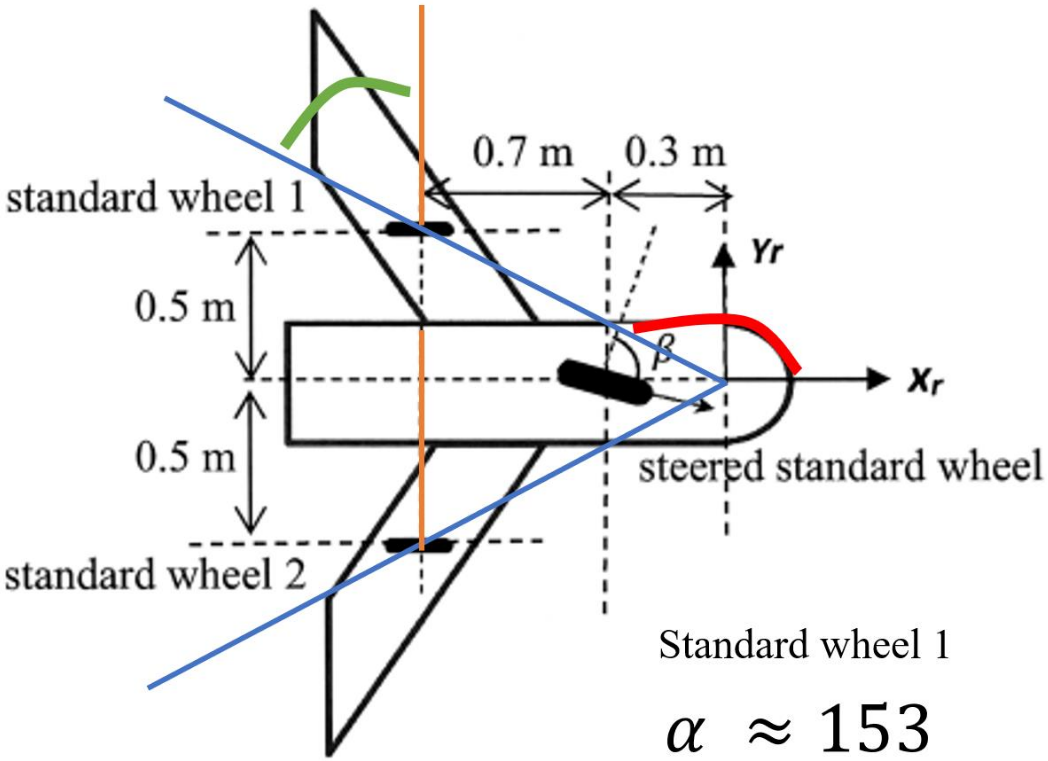
$$\beta \approx -117$$





Steered standard wheel

$$\alpha \approx 180$$
$$\beta' \approx \beta - 180$$



Standard wheel 1

$\alpha \approx 153$

$\beta \approx -63$

$$J(\beta) = [\sin(\alpha+\beta), -\cos(\alpha+\beta), -l\cos(\beta)]$$
$$C(\beta) = [\cos(\alpha+\beta), \sin(\alpha+\beta), l\sin(\beta)]$$

Standard Wheel & Steered Standard Wheel:

Rolling Constraint:

$$J(\beta)R(\theta)\dot{\xi}_I - r_{ss}\dot{\phi}_{ss} = 0$$

Sliding Constraint:

$$C(\beta)R(\theta)\dot{\xi}_I = 0$$

Caster Wheel:

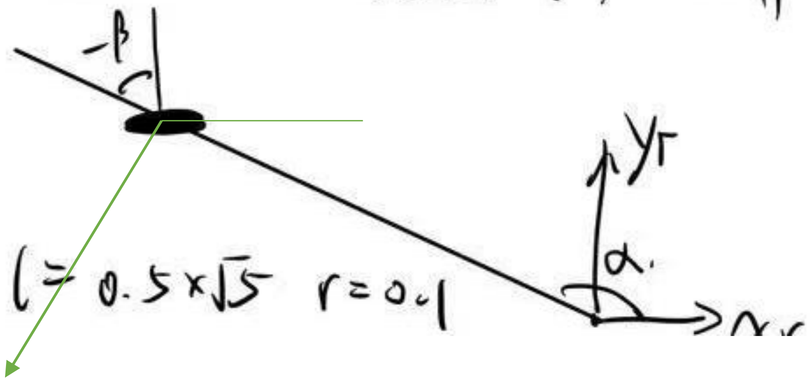
Rolling Constraint:

$$J(\beta)R(\theta)\dot{\xi}_I - r_{ss}\dot{\phi}_{ss} = 0$$

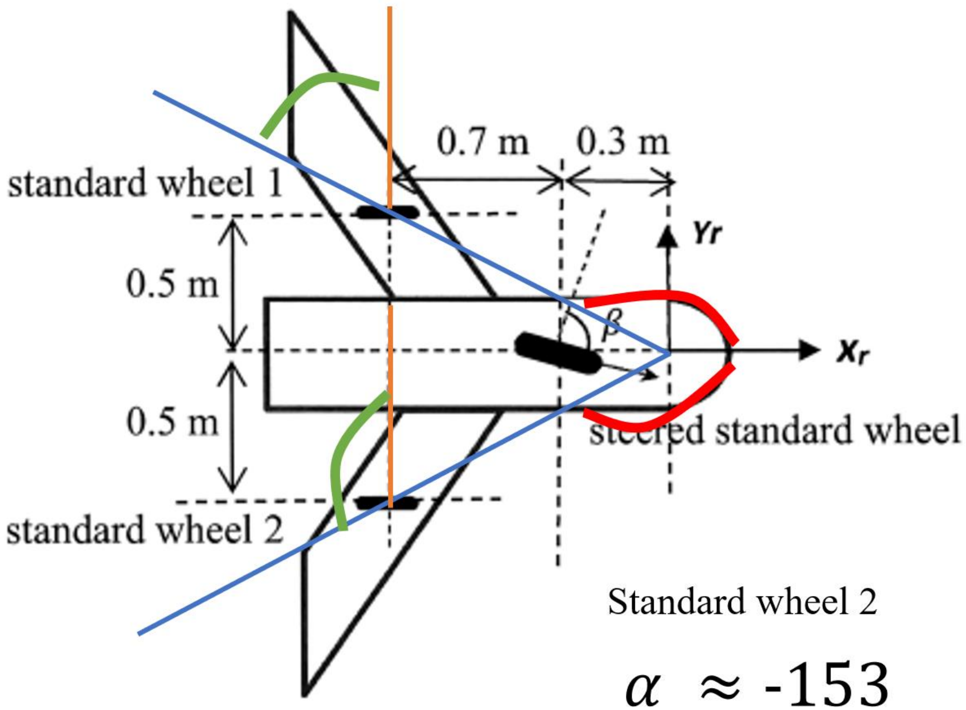
Sliding Constraint:

$$C(\beta)R(\theta)\dot{\xi}_I + d\dot{\beta} = 0$$

standard wheel 1: $\alpha+\beta = \frac{\pi}{2}$, $\cos\beta = \frac{1}{\sqrt{5}}$ $\sin\beta = -\frac{2}{\sqrt{5}}$.



$$[1,\ 0,\ -0.5]\,R(\theta)\dot{\xi} = 0.1\,\dot{\varphi}_{s_1}$$
$$[0,\ 1,\ -1]\,R(\theta)\dot{\xi} = 0$$



Standard wheel 2

$\alpha \approx -153$

$\beta \approx -117$

$$J(\beta) = [\sin(\alpha+\beta), -\cos(\alpha+\beta), -l\cos(\beta)]$$

$$C(\beta) = [\cos(\alpha+\beta), \sin(\alpha+\beta), l\sin(\beta)]$$

Standard Wheel & Steered Standard Wheel:

Rolling Constraint:

$$J(\beta)R(\theta)\dot{\xi}_I - r_{ss}\dot{\phi}_{ss} = 0$$

Sliding Constraint:

$$C(\beta)R(\theta)\dot{\xi}_I = 0$$

Caster Wheel:

Rolling Constraint:

$$J(\beta)R(\theta)\dot{\xi}_I - r_{ss}\dot{\phi}_{ss} = 0$$

Sliding Constraint:

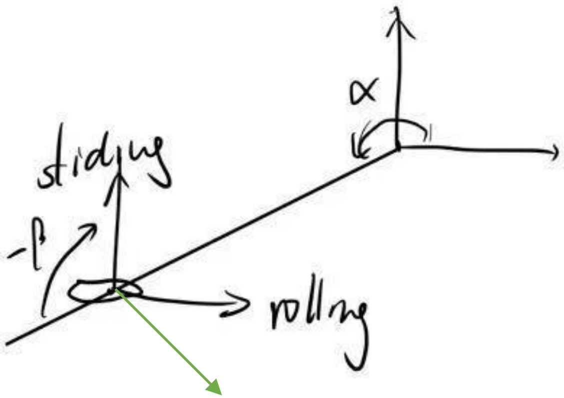
$$C(\beta)R(\theta)\dot{\xi}_I + d\dot{\beta} = 0$$

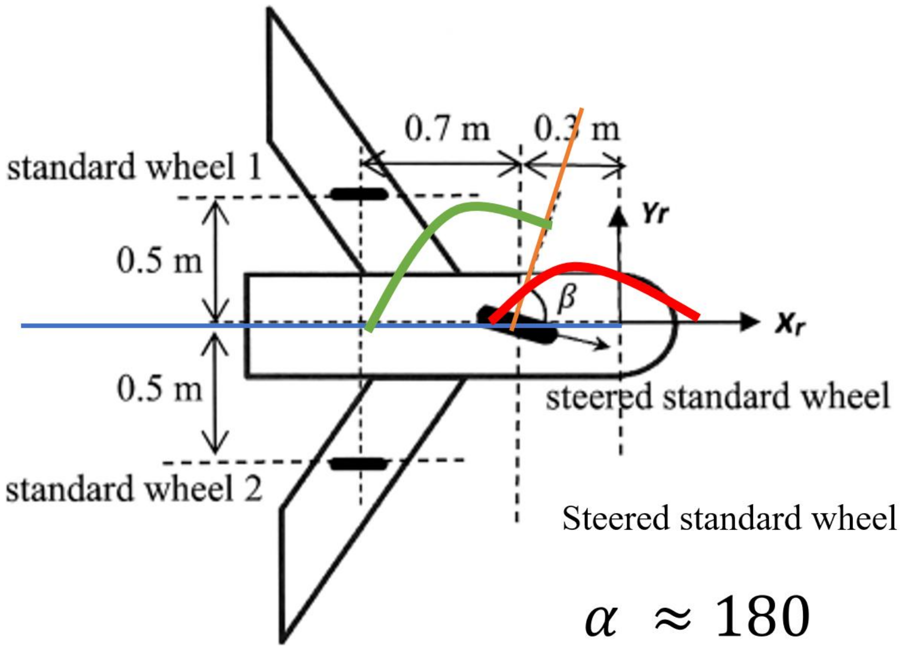
standard wheel 2 :

$\alpha+\beta = \frac{1}{2}\pi$. $\cos\beta = \frac{1}{\sqrt{5}}$ $\sin\beta = -\frac{2}{\sqrt{5}}$. $l = \frac{\sqrt{5}}{2}$ $r=0.1$

$$[1, 0, 0.5]\, R(\theta)\dot{\xi} = 0.1\,\dot{\varphi}_{s2}$$

$$[0, 1, -1]\, R(\theta)\dot{\xi} = 0$$





$\alpha \approx 180$

$\beta' \approx \beta\text{-}180$

$$J(\beta) = [\sin(\alpha+\beta), -\cos(\alpha+\beta), -l\cos(\beta)]$$
$$C(\beta) = [\cos(\alpha+\beta), \sin(\alpha+\beta), l\sin(\beta)]$$

Standard Wheel & Steered Standard Wheel:

Rolling Constraint:

$$J(\beta)R(\theta)\dot{\xi}_I - r_{ss}\dot{\phi}_{ss} = 0$$

Sliding Constraint:

$$C(\beta)R(\theta)\dot{\xi}_I = 0$$

Caster Wheel:

Rolling Constraint:

$$J(\beta)R(\theta)\dot{\xi}_I - r_{ss}\dot{\phi}_{ss} = 0$$

Sliding Constraint:

$$C(\beta)R(\theta)\dot{\xi}_I + d\dot{\beta} = 0$$

Q3: (a). steered standard wheel:

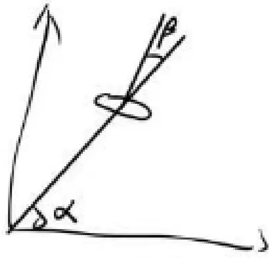
$\alpha+\beta = \beta \quad \beta = \beta - \pi \quad r = 0.1 \quad l = 0.3 \quad d = 0$

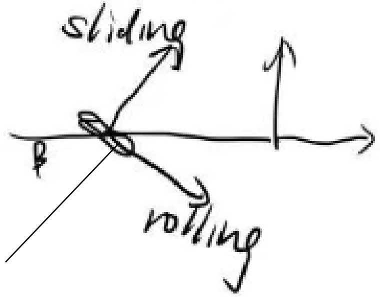
rolling:

$$[\sin\beta \quad -\cos\beta \quad 0.3\cos\beta]\, R(\theta)\, \dot{\xi}_I = 0.1\, \dot{\varphi}_{ss}$$

sliding:

$$[\cos\beta, \quad \sin\beta, \quad -0.3\sin\beta]\, R(\theta)\, \dot{\xi}_I = 0$$





Standard wheel 1

$$[1,\ 0,\ -0.5]\,R(\theta)\dot{\xi} = 0.1\,\dot{\varphi}_{s1}$$

$$[0,\ 1,\ -1]\,R(\theta)\dot{\xi} = 0$$

Steered standard wheel

rolling:

$$[\sin\beta\ \ -\cos\beta\ \ \ 0.3\cos\beta]\,R(\theta)\cdot\dot{\xi}_I = 0.1\,\dot{\varphi}_{ss}$$

sliding:

$$[\cos\beta,\ -\sin\beta,\ -0.3\sin\beta]\,R(\theta)\,\dot{\xi}_I = 0$$

Standard wheel 2

$$[1,\ 0,\ 0.5]\,R(\theta)\dot{\xi} = 0.1\,\dot{\varphi}_{s2}$$

$$[0,\ 1,\ -1]\,R(\theta)\dot{\xi} = 0$$

rolling

$$\begin{array}{l}\text{Steer}\\ \text{standard 1}\\ \text{standard 2}\end{array}\overbrace{\begin{bmatrix}\sin\beta & -\cos\beta & 0.3\cos\beta\\ 1 & 0 & -0.5\\ 1 & 0 & 0.5\end{bmatrix}}^{J}\begin{bmatrix}\dot{x}_I\\ \dot{y}_I\\ \dot{\theta}\end{bmatrix} = \overbrace{\begin{bmatrix}0.1 & 0 & 0\\ 0 & 0.1 & 0\\ 0 & 0 & 0.1\end{bmatrix}}^{r}\overbrace{\begin{bmatrix}\dot{\varphi}_{ss}\\ \dot{\varphi}_{s1}\\ \dot{\varphi}_{s2}\end{bmatrix}}^{\varphi_{ss}}$$

$$\text{sliding}\ \begin{bmatrix}\cos\beta & -\sin\beta & -0.3\sin\beta\\ 0 & 1 & -1\\ 0 & 1 & -1\end{bmatrix}\begin{bmatrix}\dot{x}_I\\ \dot{y}_I\\ \dot{\theta}\end{bmatrix} = \begin{bmatrix}0\\ 0\\ 0\end{bmatrix}$$

$$C \qquad R(\theta)\dot{\xi}$$

$$J(\beta) = [\sin(\alpha+\beta), -\cos(\alpha+\beta), -l\cos(\beta)]$$
$$C(\beta) = [\cos(\alpha+\beta), \sin(\alpha+\beta), l\sin(\beta)]$$

Parameters:

$\alpha + \beta$

β

ℓ

r

ϕ

d

Standard Wheel & Steered Standard Wheel:

Rolling Constraint:

$$J(\beta)R(\theta)\dot{\xi}_I - r_{ss}\dot{\phi}_{ss} = 0$$

Sliding Constraint:

$$C(\beta)R(\theta)\dot{\xi}_I = 0$$

Caster Wheel:

Rolling Constraint:

$$J(\beta)R(\theta)\dot{\xi}_I - r_{ss}\dot{\phi}_{ss} = 0$$

Sliding Constraint:

$$C(\beta)R(\theta)\dot{\xi}_I + d\dot{\beta} = 0$$

rolling

$$\begin{array}{l} \text{Steer} \\ \text{standard } 1 \\ \text{standard } 2 \end{array} \overbrace{\begin{bmatrix} \sin\beta & -\cos\beta & 0.3\cos\beta \\ 1 & 0 & -0.5 \\ 1 & 0 & 0.5 \end{bmatrix}}^{J} \begin{bmatrix} \dot{x}_I \\ \dot{y}_I \\ \dot{\theta} \end{bmatrix} = \overbrace{\begin{bmatrix} 0.1 & 0 & 0 \\ 0 & 0.1 & 0 \\ 0 & 0 & 0.1 \end{bmatrix}}^{r} \overbrace{\begin{bmatrix} \dot{\varphi}_{ss} \\ \dot{\varphi}_{s1} \\ \dot{\varphi}_{s2} \end{bmatrix}}^{\dot{\phi}_{ss}}$$

sliding

$$\underbrace{\begin{bmatrix} \cos\beta & -\sin\beta & -0.3\sin\beta \\ 0 & 1 & -1 \\ 0 & 1 & -1 \end{bmatrix}}_{C} \underbrace{\begin{bmatrix} \dot{x}_I \\ \dot{y}_I \\ \dot{\theta} \end{bmatrix}}_{R(\theta)\xi} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

24-25 S2 Q3

A three-wheeled mobile robot with a circular platform is shown in Figure 2. The mobile has two identical standard wheels with a radius of 0.090 m and one castor wheel with a radius of 0.025 m. A local reference frame (x_r, y_r) is assigned as shown in the figure. The rotational velocities of the castor wheel, standard wheel 1 and standard wheel 2 are denoted by $\dot{\phi}_c$, $\dot{\phi}_{s1}$ and $\dot{\phi}_{s2}$ respectively and the velocity of the steering angle of the castor wheel is denoted by $\dot{\beta}$. Derive the rolling and sliding constraints of the mobile robot.

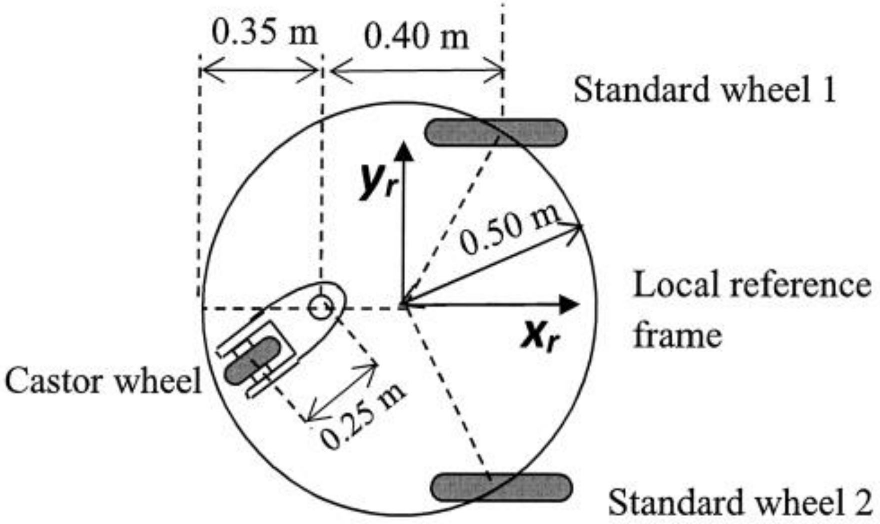
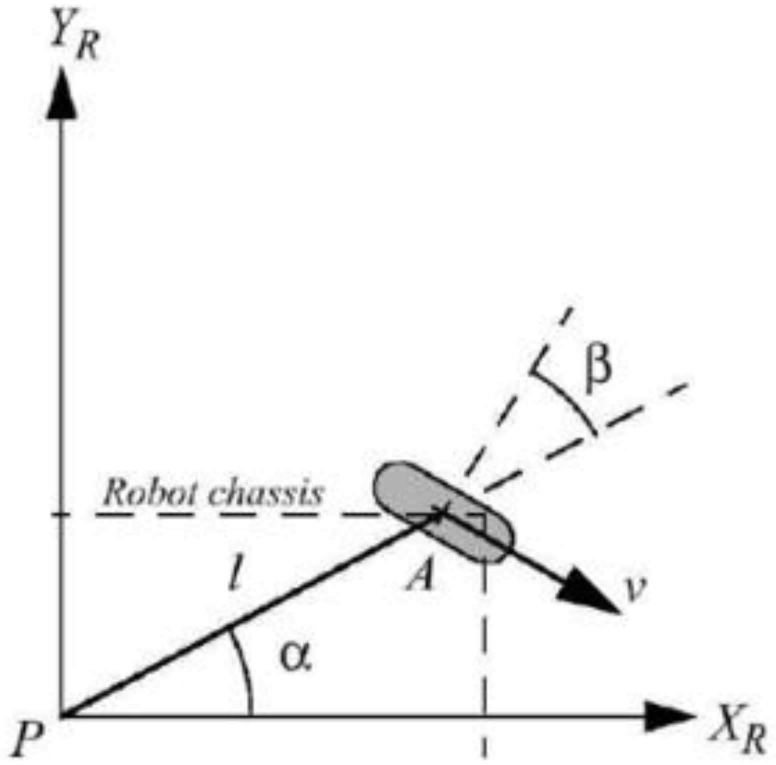


Figure 2



(8 Marks)

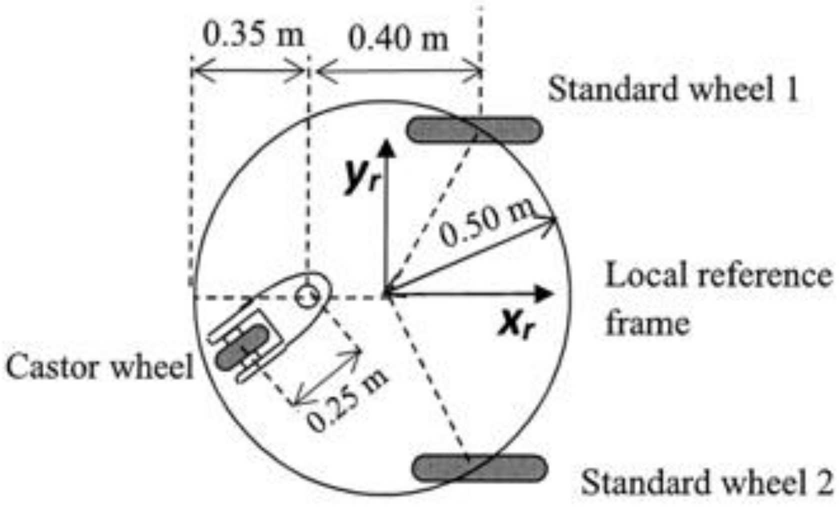
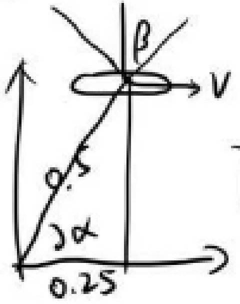
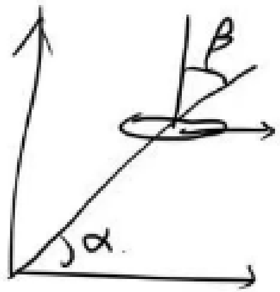


Figure 2

Standard wheel 1:





$\alpha = \frac{\pi}{3}$ $\beta = \frac{\pi}{6}$. $l = 0.5$ $d = 0$. $r = 0.09$

$[1, \ 0, \ -\frac{\sqrt{3}}{4}] R(\theta) \dot{\xi} = 0.09 \dot{\varphi}_{s1}$

$[0, \ 1, \ \frac{1}{4}] R(\theta) \dot{\xi} = 0.$

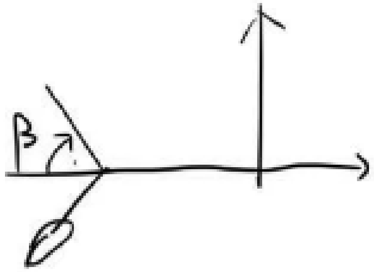
Standard wheel 2:

$\alpha = -\frac{\pi}{3}$ $\beta = -\frac{\pi}{6} + \pi$ $l = 0.5$ $r = 0.09$

$[1, \ 0, \ \frac{\sqrt{3}}{4}] R(\theta) \dot{\xi} = 0.09 \dot{\varphi}_{s2}$

$[0, \ 1, \ \frac{1}{4}] R(\theta) \dot{\xi} = 0.$

Castor wheel: $\alpha = \pi$ $\beta = -\beta(t)$ $r = 0.025$ $l = 0.15$ $d = 0.25$



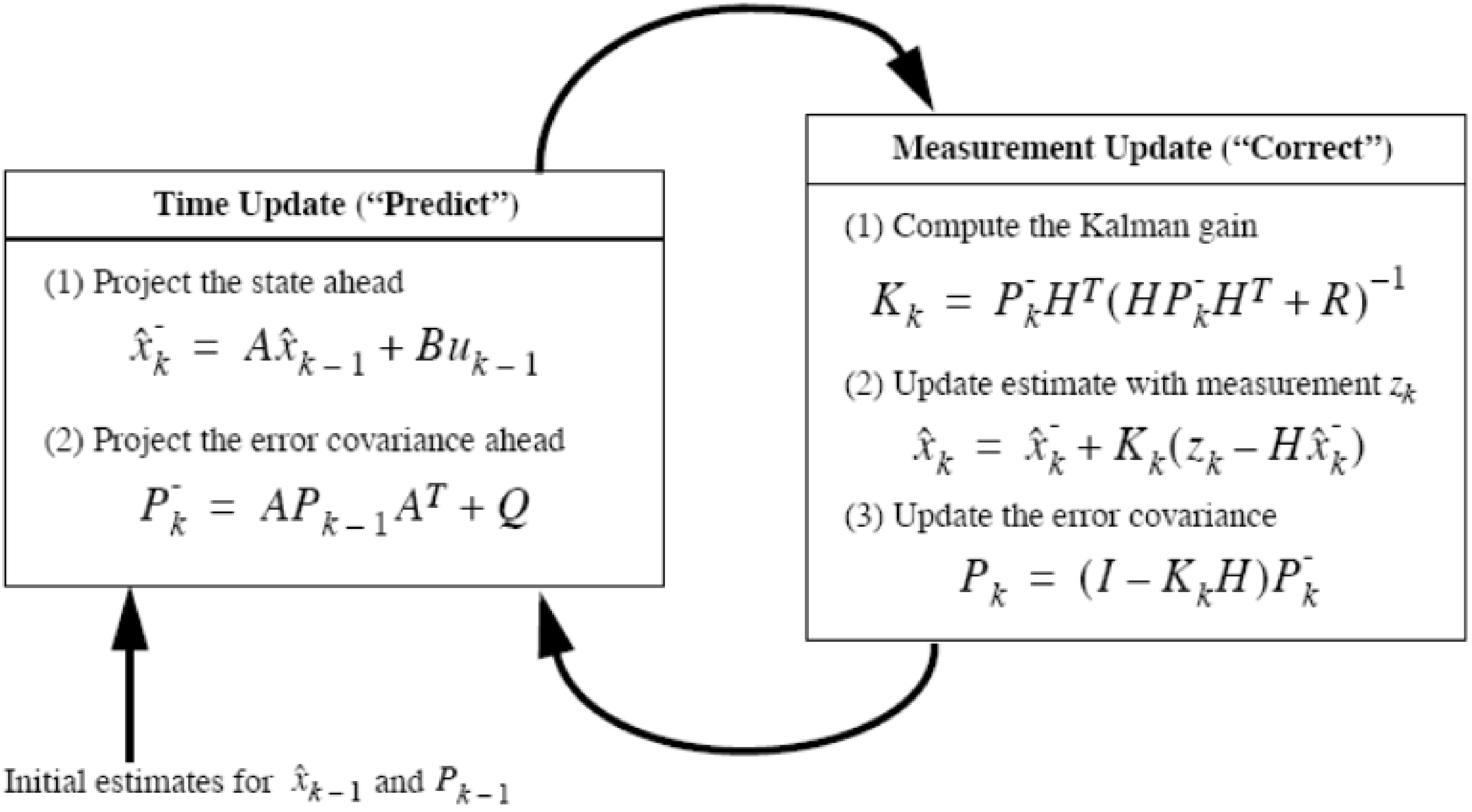
$[\sin\beta, \ \cos\beta, -0.15\cos\beta] R(\theta) \dot{\xi} = 0.025 \dot{\varphi}_c$

$[-\cos\beta, \ \sin\beta, \ -0.15 \sin\beta] R(\theta) \dot{\xi} + 0.25 \dot{\beta} = 0$

Kalman Filter

Sequential State Estimation

Sequential State Estimation by Kalman Filtering



5. Two sensors are used to measure a state variable x_k that can be modelled by $x_{k+1} = x_k$. The measurements of the two sensors are given by y_{1k} and y_{2k}, respectively, which are governed by the following models:

$$y_{1k} = x_k + v_{1k},\ y_{2k} = x_k + v_{2k}$$

where v_{1k} and v_{2k} are zero mean Gaussian sensor noises with variance given by a^2 and $9a^2$, respectively.

Let $\hat{x}_k$ represent the estimate of x_k and $\hat{x}_{k+1}$ represent the estimate of x_{k+1}. Let the estimation errors be $\tilde{x}_k = x_k - \hat{x}_k$ and $\tilde{x}_{k+1} = x_{k+1} - \hat{x}_{k+1}$. Assume that the estimation error $\tilde{x}_k$ and the noise terms v_{1k} and v_{2k} are uncorrelated. Assume that $E[\tilde{x}_k] = 0$ and $E[\tilde{x}_{k+1}] = 0$. Let the estimation error variances be $p_k = E[\tilde{x}_k^2]$ and $p_{k+1} = E[\tilde{x}_{k+1}^2]$, respectively.

Two estimation strategies are designed as follows:

Strategy 1:

$$\hat{x}_{k+1} = \hat{x}_k + L_{ck}(y_{1k} - \hat{x}_k) + L_{ck}(y_{2k} - \hat{x}_k)$$

Strategy 2:

$$\hat{x}_{k+1} = \hat{x}_k + L_{1k}(y_{1k} - \hat{x}_k) + L_{2k}(y_{2k} - \hat{x}_k)$$

where L_{ck}, L_{1k}, and L_{2k} represent respective Kalman gains.

Define Estimation Error:

$$\tilde{x}_{k+1} = x_{k+1} - \hat{x}_{k+1}$$

Calculate Variance:

$$P_{k+1} = \mathrm{E}[\tilde{x}_{k+1}]^2$$

Minimize Variance:

$$\frac{\partial p_{k+1}}{\partial L_{ck}} = 0$$

(a) Design the update laws for the Kaman gains L_{ck}, L_{1k} and L_{2k} in order to minimize the corresponding estimation error variance.

(12 Marks)

Define Estimation Error:

$$\tilde{x}_{k+1} = x_{k+1} - \hat{x}_{k+1}$$

Calculate Variance:

$$P_{k+1} = \mathrm{E}[\tilde{x}_{k+1}]^2$$

Minimize Variance:

$$\frac{\partial p_{k+1}}{\partial L_{ck}} = 0$$

For strategy 1

5. Two sensors are used to measure a state variable x_k that can be modelled by $x_{k+1} = x_k$. The measurements of the two sensors are given by y_{1k} and y_{2k}, respectively, which are governed by the following models:

$$y_{1k} = x_k + v_{1k},\ y_{2k} = x_k + v_{2k}$$

where v_{1k} and v_{2k} are zero mean Gaussian sensor noises with variance given by a^2 and $9a^2$, respectively.

Let $\hat{x}_k$ represent the estimate of x_k and $\hat{x}_{k+1}$ represent the estimate of x_{k+1}. Let the estimation errors be $\tilde{x}_k = x_k - \hat{x}_k$ and $\tilde{x}_{k+1} = x_{k+1} - \hat{x}_{k+1}$. Assume that the estimation error $\tilde{x}_k$ and the noise terms v_{1k} and v_{2k} are uncorrelated. Assume that $E[\tilde{x}_k] = 0$ and $E[\tilde{x}_{k+1}] = 0$. Let the estimation error variances be $p_k = E[\tilde{x}_k^2]$ and $p_{k+1} = E[\tilde{x}_{k+1}^2]$, respectively.

Two estimation strategies are designed as follows:
Strategy 1:

$$\hat{x}_{k+1} = \hat{x}_k + L_{ck}(y_{1k} - \hat{x}_k) + L_{ck}(y_{2k} - \hat{x}_k)$$

Define Estimation Error:

$$\tilde{x}_{k+1} = x_{k+1} - \hat{x}_{k+1}$$

Since $x_{k+1} = x_k$:

$$\tilde{x}_{k+1} = x_k - [\hat{x}_k + L_{ck}(y_{1k} - \hat{x}_k) + L_{ck}(y_{2k} - \hat{x}_k)]$$

$$\tilde{x}_{k+1} = (x_k - \hat{x}_k) - L_{ck}(x_k + v_{1k} - \hat{x}_k) - L_{ck}(x_k + v_{2k} - \hat{x}_k)$$

$$\tilde{x}_{k+1} = \tilde{x}_k - L_{ck}(\tilde{x}_k + v_{1k}) - L_{ck}(\tilde{x}_k + v_{2k})$$

$$\tilde{x}_{k+1} = (1 - 2L_{ck})\tilde{x}_k - L_{ck}v_{1k} - L_{ck}v_{2k}$$

5. Two sensors are used to measure a state variable x_k that can be modelled by $x_{k+1} = x_k$. The measurements of the two sensors are given by y_{1k} and y_{2k}, respectively, which are governed by the following models:

$$y_{1k} = x_k + v_{1k},\ y_{2k} = x_k + v_{2k}$$

where v_{1k} and v_{2k} are zero mean Gaussian sensor noises with variance given by a^2 and $9a^2$, respectively.

Let $\hat{x}_k$ represent the estimate of x_k and $\hat{x}_{k+1}$ represent the estimate of x_{k+1}. Let the estimation errors be $\tilde{x}_k = x_k - \hat{x}_k$ and $\tilde{x}_{k+1} = x_{k+1} - \hat{x}_{k+1}$. Assume that the estimation error $\tilde{x}_k$ and the noise terms v_{1k} and v_{2k} are uncorrelated. Assume that $E[\tilde{x}_k] = 0$ and $E[\tilde{x}_{k+1}] = 0$. Let the estimation error variances be $p_k = E[\tilde{x}_k^2]$ and $p_{k+1} = E[\tilde{x}_{k+1}^2]$, respectively.

Two estimation strategies are designed as follows:

Strategy 1:

$$\hat{x}_{k+1} = \hat{x}_k + L_{ck}(y_{1k} - \hat{x}_k) + L_{ck}(y_{2k} - \hat{x}_k)$$

Variance

$$V[X] = E[(X - E[X])^2] \\ = E[X^2] - (E[X])^2$$

$V(X) = E[X^2]$ `for zero-mean X`

$$\tilde{x}_{k+1} = (1 - 2L_{ck})\tilde{x}_k - L_{ck}v_{1k} - L_{ck}v_{2k}$$

Calculate Variance:

$$p_{k+1} = E[\tilde{x}_{k+1}^2] = E[((1 - 2L_{ck})\tilde{x}_k - L_{ck}v_{1k} - L_{ck}v_{2k})^2]$$

Since $\tilde{x}_k, v_{1k}, v_{2k}$ **are uncorrelated and zero mean:**

$$p_{k+1} = (1 - 2L_{ck})^2 E[\tilde{x}_k^2] + L_{ck}^2 E[v_{1k}^2] + L_{ck}^2 E[v_{2k}^2]$$

$$p_{k+1} = (1 - 2L_{ck})^2 p_k + L_{ck}^2\sigma_1^2 + L_{ck}^2\sigma_2^2$$

$$p_{k+1} = (1 - 2L_{ck})^2 p_k + L_{ck}^2(a^2 + 9a^2)$$ [cite: 175]

$$p_{k+1} = (1 - 4L_{ck} + 4L_{ck}^2)p_k + 10a^2L_{ck}^2$$

$$p_{k+1} = (1 - 4L_{ck} + 4L_{ck}^2)p_k + 10a^2L_{ck}^2$$

Minimize Variance: Set the derivative with respect to L_{ck} to zero.

$$\frac{\partial p_{k+1}}{\partial L_{ck}} = (-4 + 8L_{ck})p_k + 20a^2L_{ck} = 0$$

$$-4p_k + 8L_{ck}p_k + 20a^2L_{ck} = 0$$

$$L_{ck}(8p_k + 20a^2) = 4p_k$$

$$L_{ck} = \frac{4p_k}{8p_k + 20a^2} = \boxed{\frac{p_k}{2p_k + 5a^2}}$$

For strategy 2

5. Two sensors are used to measure a state variable x_k that can be modelled by $x_{k+1} = x_k$. The measurements of the two sensors are given by y_{1k} and y_{2k}, respectively, which are governed by the following models:

$$y_{1k} = x_k + v_{1k}, \; y_{2k} = x_k + v_{2k}$$

where v_{1k} and v_{2k} are zero mean Gaussian sensor noises with variance given by a^2 and $9a^2$, respectively.

Let $\hat{x}_k$ represent the estimate of x_k and $\hat{x}_{k+1}$ represent the estimate of x_{k+1}. Let the estimation errors be $\tilde{x}_k = x_k - \hat{x}_k$ and $\tilde{x}_{k+1} = x_{k+1} - \hat{x}_{k+1}$. Assume that the estimation error $\tilde{x}_k$ and the noise terms v_{1k} and v_{2k} are uncorrelated. Assume that $E[\tilde{x}_k] = 0$ and $E[\tilde{x}_{k+1}] = 0$. Let the estimation error variances be $p_k = E\left[\tilde{x}_k^2\right]$ and $p_{k+1} = E\left[\tilde{x}_{k+1}^2\right]$, respectively.

Two estimation strategies are designed as follows:

Strategy 1:

$$\hat{x}_{k+1} = \hat{x}_k + L_{ck}(y_{1k} - \hat{x}_k) + L_{ck}(y_{2k} - \hat{x}_k)$$

Strategy 2:

$$\hat{x}_{k+1} = \hat{x}_k + L_{1k}(y_{1k} - \hat{x}_k) + L_{2k}(y_{2k} - \hat{x}_k)$$

Define Estimation Error: [cite: 65

$$\tilde{x}_{k+1} = x_{k+1} - \hat{x}_{k+1}$$
$$\tilde{x}_{k+1} = x_k - [\hat{x}_k + L_{1k}(y_{1k} - \hat{x}_k) + L_{2k}(y_{2k} - \hat{x}_k)]$$
$$\tilde{x}_{k+1} = \tilde{x}_k - L_{1k}(\tilde{x}_k + v_{1k}) - L_{2k}(\tilde{x}_k + v_{2k})$$
$$\tilde{x}_{k+1} = (1 - L_{1k} - L_{2k})\tilde{x}_k - L_{1k}v_{1k} - L_{2k}v_{2k}$$

Variance

$$V[X] = E\left[(X - E[X])^2\right] = E\left[X^2\right] - (E[X])^2$$

$V(X) = E\left[X^2\right]$ `for zero-mean X`

5. Two sensors are used to measure a state variable x_k that can be modelled by $x_{k+1} = x_k$. The measurements of the two sensors are given by y_{1k} and y_{2k}, respectively, which are governed by the following models:

$$y_{1k} = x_k + v_{1k},\ y_{2k} = x_k + v_{2k}$$

where v_{1k} and v_{2k} are zero mean Gaussian sensor noises with variance given by a^2 and $9a^2$, respectively.

Let $\hat{x}_k$ represent the estimate of x_k and $\hat{x}_{k+1}$ represent the estimate of x_{k+1}. Let the estimation errors be $\tilde{x}_k = x_k - \hat{x}_k$ and $\tilde{x}_{k+1} = x_{k+1} - \hat{x}_{k+1}$. Assume that the estimation error $\tilde{x}_k$ and the noise terms v_{1k} and v_{2k} are uncorrelated. Assume that $E\left[\tilde{x}_k\right] = 0$ and $E\left[\tilde{x}_{k+1}\right] = 0$. Let the estimation error variances be $p_k = E\left[\tilde{x}_k^2\right]$ and $p_{k+1} = E\left[\tilde{x}_{k+1}^2\right]$, respectively.

Two estimation strategies are designed as follows:
Strategy 1:

$$\hat{x}_{k+1} = \hat{x}_k + L_{1k}\left(y_{1k} - \hat{x}_k\right) + L_{2k}\left(y_{2k} - \hat{x}_k\right)$$

$$\tilde{x}_{k+1} = (1 - L_{1k} - L_{2k})\tilde{x}_k - L_{1k}v_{1k} - L_{2k}v_{2k}$$

Calculate Variance: [cite: 66]

$$p_{k+1} = E[\tilde{x}_{k+1}^2] = E[((1 - L_{1k} - L_{2k})\tilde{x}_k - L_{1k}v_{1k} - L_{2k}v_{2k})^2]$$
$$p_{k+1} = (1 - L_{1k} - L_{2k})^2 E[\tilde{x}_k^2] + L_{1k}^2 E[v_{1k}^2] + L_{2k}^2 E[v_{2k}^2]$$
$$p_{k+1} = (1 - L_{1k} - L_{2k})^2 p_k + L_{1k}^2 \sigma_1^2 + L_{2k}^2 \sigma_2^2$$
$$p_{k+1} = (1 - L_{1k} - L_{2k})^2 p_k + L_{1k}^2 a^2 + L_{2k}^2 (9a^2)$$ [cite: 175]

Minimize Variance: Set partial derivatives with respect to L_{1k} and L_{2k} to zero

- $\frac{\partial p_{k+1}}{\partial L_{1k}} = 2(1 - L_{1k} - L_{2k})(-1)p_k + 2L_{1k}a^2 = 0$ [cite: 68]
  $-(1 - L_{1k} - L_{2k})p_k + L_{1k}a^2 = 0$ (1)
- $\frac{\partial p_{k+1}}{\partial L_{2k}} = 2(1 - L_{1k} - L_{2k})(-1)p_k + 2L_{2k}(9a^2) = 0$ [cite: 69]
  $-(1 - L_{1k} - L_{2k})p_k + 9a^2L_{2k} = 0$ (2)

$$\begin{cases} -(1 - L_{1k} - L_{2k})p_k + L_{1k}a^2 = 0 \\ -(1 - L_{1k} - L_{2k})p_k + 9a^2L_{2k} = 0 \end{cases}$$

$$L_{2k} = \frac{p_k}{10p_k + 9a^2}$$

$$L_{1k} = 9L_{2k} = \frac{9p_k}{10p_k + 9a^2}$$

$$p_{k+1} = (1 - L_{1k} - L_{2k})^2 p_k + L_{1k}^2 a^2 + L_{2k}^2 (9a^2)$$

$$L_{2k} = \frac{p_k}{10p_k + 9a^2}$$

$$L_{1k} = 9L_{2k} = \frac{9p_k}{10p_k + 9a^2}$$

$$K_k = \frac{HAP_k}{H^2 P_k + R}$$

$$P_{k+1} = -\frac{(AHP_k)^2}{H^2 P_k + R} + Q + A^2 P_k$$

$$p_{k+1} = (1 - 4L_{ck} + 4L_{ck}^2) p_k + 10a^2 L_{ck}^2$$

$$L_{ck} = \frac{4p_k}{8p_k + 20a^2} = \frac{p_k}{2p_k + 5a^2}$$

(b) Compare the two strategies and discuss the advantages and disadvantages of each strategy. Justify your answer in detail.

(8 Marks)

Batch Processing for each sensor VS Single Gain for each sensor

Smaller possible error variance
More computation resource

$$p_{k+1} = (1 - L_{1k} - L_{2k})^2 p_k + L_{1k}^2 a^2 + L_{2k}^2 (9a^2)$$

$$L_{2k} = \frac{p_k}{10p_k + 9a^2}$$

$$L_{1k} = 9L_{2k} = \frac{9p_k}{10p_k + 9a^2}$$

$$P_{k+1} = \frac{9a^2 P_k}{10P_k + 9a^2}$$

$$p_{k+1} = (1 - 4L_{ck} + 4L_{ck}^2)p_k + 10a^2 L_{ck}^2$$

$$L_{ck} = \frac{4p_k}{8p_k + 20a^2} = \frac{p_k}{2p_k + 5a^2}$$

$$P_{k+1} = \frac{5a^2 P_k}{2P_k + 5a^2}$$

$$\frac{9a^2 P_k}{10P_k + 9a^2} < \frac{5a^2 P_k}{2P_k + 5a^2}$$

24-25 S2 Q5

Define Estimation Error:

$$\tilde{x}_{k+1} = x_{k+1} - \hat{x}_{k+1}$$

Calculate Variance:

$$P_{k+1} = \mathrm{E}[\tilde{x}_{k+1}]^2$$

Minimize Variance:

$$\frac{\partial p_{k+1}}{\partial L_{ck}} = 0$$

Four sensors are available to measure a state variable x_k that can be modelled by $x_{k+1} = x_k$. The measurements of the four sensors are given by y_{1k}, y_{2k}, y_{3k}, and y_{4k}, respectively, which are governed by the models $y_{1k} = x_k + v_{1k}$, $y_{2k} = x_k + v_{2k}$, $y_{3k} = x_k + v_{3k}$ and $y_{4k} = x_k + v_{4k}$ where v_{1k}, v_{2k}, v_{3k}, and v_{4k} are zero mean Gaussian sensor noises with variance given by a^2, $(2a)^2$, $(3a)^2$, and $(4a)^2$, respectively.

Let $\hat{x}_k$ represent the estimate of x_k and $\hat{x}_{k+1}$ represent the estimate of x_{k+1}. Let the estimation errors be $\tilde{x}_k = x_k - \hat{x}_k$ and $\tilde{x}_{k+1} = x_{k+1} - \hat{x}_{k+1}$. Assume that the estimation error $\tilde{x}_k$ and the noise terms v_{1k}, v_{2k}, v_{3k}, and v_{4k} are uncorrelated. Assume that $E[\tilde{x}_k] = 0$ and $E[\tilde{x}_{k+1}] = 0$. Let the estimation error variances be $p_k = E[\tilde{x}_k^2]$ and $p_{k+1} = E[\tilde{x}_{k+1}^2]$.

(a) Choose sensors 1 and 4 for estimation. Design the estimation strategy 1 as follows:

$$\hat{x}_{k+1} = \hat{x}_k + L_{1k}(y_{1k} - \hat{x}_k) + L_{4k}(y_{4k} - \hat{x}_k)$$

where L_{1k} and L_{4k} represent Kalman gains. Design the update laws for the Kaman gains to minimize the estimation error variance.

(7 Marks)

(b) Choose sensors 2 and 3 for estimation. Design the estimation strategy 2 as follows:

$$\hat{x}_{k+1} = \hat{x}_k + L_{2k}(y_{2k} - \hat{x}_k) + L_{3k}(y_{3k} - \hat{x}_k)$$

Three sensors are used to measure a state variable x_k that can be modelled by $x_{k+1} = x_k$. The measurements of the three sensors are given by y_{1k}, y_{2k}, and y_{3k}, respectively, which are governed by the models $y_{1k} = x_k + v_{1k}$, $y_{2k} = x_k + v_{2k}$, and $y_{3k} = x_k + v_{3k}$ where v_{1k}, v_{2k}, and v_{3k} are zero mean Gaussian sensor noises with variance given by a^2, $4a^2$, and $9a^2$, respectively.

Let $\hat{x}_k$ represent the estimate of x_k and $\hat{x}_{k+1}$ represent the estimate of x_{k+1}. Let the estimation errors be $\tilde{x}_k = x_k - \hat{x}_k$ and $\tilde{x}_{k+1} = x_{k+1} - \hat{x}_{k+1}$. Assume that the estimation error $\tilde{x}_k$ and the noise terms v_{1k}, v_{2k}, and v_{3k} are uncorrelated. Assume that $E[\tilde{x}_k] = 0$ and $E[\tilde{x}_{k+1}] = 0$. Let the estimation error variances be $p_k = E[\tilde{x}_k^2]$ and $p_{k+1} = E[\tilde{x}_{k+1}^2]$.

(a) Design an estimation strategy as follows:

$$\hat{x}_{k+1} = \hat{x}_k + L_{1k}(y_{1k} - \hat{x}_k) + L_{2k}(y_{2k} - \hat{x}_k) + L_{3k}(y_{3k} - \hat{x}_k)$$

where L_{1k}, L_{2k}, and L_{3k} represent Kalman gains. Design the update laws for the Kalman gains to minimize the estimation error variance.

(10 Marks)

(b) Design an estimation strategy using one Kalman gain for the three sensors. Compare this estimation strategy to the estimation strategy in part (a) and discuss the advantages and disadvantages of each strategy. Justify your answer.

(10 Marks)

(a) Design an estimation strategy as follows:

$$\hat{x}_{k+1} = \hat{x}_k + L_{1k}(y_{1k} - \hat{x}_k) + L_{2k}(y_{2k} - \hat{x}_k) + L_{3k}(y_{3k} - \hat{x}_k)$$

where L_{1k}, L_{2k}, L_{3k} represents Kalman gains.

Since $y_{ik} = x_k + v_{ik}$, we have $y_{ik} - \hat{x}_k = x_k - \hat{x}_k + v_{ik} = \tilde{x}_k + v_{ik}$.

Since $x_{k+1} = x_k$, we have $\tilde{x}_{k+1} = x_{k+1} - \hat{x}_{k+1} = x_k - \hat{x}_{k+1}$.

Therefore,

$$\begin{aligned}
\hat{x}_{k+1} &= \hat{x}_k + L_{1k}(y_{1k} - \hat{x}_k) + L_{2k}(y_{2k} - \hat{x}_k) + L_{3k}(y_{3k} - \hat{x}_k) \\
&= \hat{x}_k + (L_{1k} + L_{2k} + L_{3k})\tilde{x}_k + L_{1k}v_{1k} + L_{2k}v_{2k} + L_{3k}v_{3k} \\
\tilde{x}_{k+1} &= x_k - \hat{x}_{k+1} \\
&= (1 - L_{1k} - L_{2k} - L_{3k})\tilde{x}_k - (L_{1k}v_{1k} + L_{2k}v_{2k} + L_{3k}v_{3k})
\end{aligned}$$

$$\begin{aligned}
p_{k+1} &= \mathbb{E}[\tilde{x}_{k+1}^2] \\
&= (1 - L_{1k} - L_{2k} - L_{3k})^2\mathbb{E}[\tilde{x}_k^2] + L_{1k}^2\mathbb{E}[v_{1k}^2] + L_{2k}^2\mathbb{E}[v_{2k}^2] + L_{3k}^2\mathbb{E}[v_{3k}^2] \\
&= (1 - L_{1k} - L_{2k} - L_{3k})^2 p_k + L_{1k}^2 \cdot a^2 + L_{2k}^2 \cdot 4a^2 + L_{3k}^2 \cdot 9a^2
\end{aligned}$$

(Use $\mathbb{E}[\tilde{x}_k] = 0, \mathbb{E}[v_{ik}] = 0$ and uncorrelated)

Define Estimation Error:

$$\tilde{x}_{k+1} = x_{k+1} - \hat{x}_{k+1}$$

Calculate Variance:

$$P_{k+1} = \mathrm{E}[\tilde{x}_{k+1}]^2$$

Minimize Variance:

$$\frac{\partial p_{k+1}}{\partial L_{ck}} = 0$$

$$
\begin{aligned}
p_{k+1} &= \mathbb{E}[\tilde{x}_{k+1}^2] \\
&= (1 - L_{1k} - L_{2k} - L_{3k})^2 \mathbb{E}[\tilde{x}_k^2] + L_{1k}^2 \mathbb{E}[v_{1k}^2] + L_{2k}^2 \mathbb{E}[v_{2k}^2] + L_{3k}^2 \mathbb{E}[v_{3k}^2] \\
&= (1 - L_{1k} - L_{2k} - L_{3k})^2 p_k + L_{1k}^2 \cdot a^2 + L_{2k}^2 \cdot 4a^2 + L_{3k}^2 \cdot 9a^2
\end{aligned}
$$

$$
\begin{cases}
\dfrac{\partial p_{k+1}}{\partial L_{1k}} = 0 \\
\dfrac{\partial p_{k+1}}{\partial L_{2k}} = 0 \\
\dfrac{\partial p_{k+1}}{\partial L_{3k}} = 0
\end{cases}
\Longrightarrow
\boxed{
\begin{cases}
L_{1k} = \dfrac{36 p_k}{36a^2 + 49 p_k} \\
L_{2k} = \dfrac{9 p_k}{36a^2 + 49 p_k} \\
L_{3k} = \dfrac{4 p_k}{36a^2 + 49 p_k}
\end{cases}
, p_{k+1} = \dfrac{36a^2 p_k}{36a^2 + 49 p_k}
}
$$

(b) Design an estimation strategy using one Kalman gain for the three sensors.

Estimation Strategy:

$$\hat{x}_{k+1} = \hat{x}_k + L_{ck}(y_{1k} + y_{2k} + y_{3k} - 3\hat{x}_k)$$

$$\begin{aligned}\tilde{x}_{k+1} &= \tilde{x}_k - L_{ck}(3\tilde{x}_k + v_{1k} + +v_{2k} + v_{3k}) \\ &= (1 - 3L_{ck})\tilde{x}_k - L_{ck}(v_{1k} + +v_{2k} + v_{3k})\end{aligned}$$

$$p_{k+1} = \mathbb{E}[\tilde{x}_{k+1}^2] = (1 - 3L_{ck})p_k + L_{ck}^2 \cdot 14a^2$$

$$\frac{\partial p_{k+1}}{\partial L_{ck}} = 0 \implies L_{ck} = \frac{3p_k}{9p_k + 14a^2}, p_{k+1} = \frac{14a^2 p_k}{9p_k + 14a^2}$$

Three sensors are used to measure a state variable x_k that can be modelled by $x_{k+1} = x_k$. The measurements of the three sensors are given by y_{1k}, y_{2k}, and y_{3k}, respectively, which are governed by the models $y_{1k} = x_k + v_{1k}$, $y_{2k} = x_k + v_{2k}$, and $y_{3k} = x_k + v_{3k}$ where v_{1k}, v_{2k}, and v_{3k} are zero mean Gaussian sensor noises with variance given by a^2, $4a^2$, and $9a^2$, respectively.

Let $\hat{x}_k$ represent the estimate of x_k and $\hat{x}_{k+1}$ represent the estimate of x_{k+1}. Let the estimation errors be $\tilde{x}_k = x_k - \hat{x}_k$ and $\tilde{x}_{k+1} = x_{k+1} - \hat{x}_{k+1}$. Assume that the estimation error $\tilde{x}_k$ and the noise terms v_{1k}, v_{2k}, and v_{3k} are uncorrelated. Assume that $E[\tilde{x}_k] = 0$ and $E[\tilde{x}_{k+1}] = 0$. Let the estimation error variances be $p_k = E[\tilde{x}_k^2]$ and $p_{k+1} = E[\tilde{x}_{k+1}^2]$.

(a) Design an estimation strategy as follows:

$$\hat{x}_{k+1} = \hat{x}_k + L_{1k}(y_{1k} - \hat{x}_k) + L_{2k}(y_{2k} - \hat{x}_k) + L_{3k}(y_{3k} - \hat{x}_k)$$

where L_{1k}, L_{2k}, and L_{3k} represent Kalman gains. Design the update laws for the Kalman gains to minimize the estimation error variance.

(10 Marks)

(b) Design an estimation strategy using one Kalman gain for the three sensors. Compare this estimation strategy to the estimation strategy in part (a) and discuss the advantages and disadvantages of each strategy. Justify your answer.

(10 Marks)

Structure and Pose Estimation

Difference between E and H

	Essential Matrix	Homography Matrix
points	8	4
requirements	no coplanar	coplanar, non-collinear
translation	nonzero translation	NA
outcomes	unique	depend on experience
accuracy	NA	more accurate

Homography matrix and 4-point

Process:

$$\alpha_j = \frac{z_j^*}{z_j}$$

$$m_j = \alpha_j \mathbf{H} m_j^*$$

$$M_j = \begin{bmatrix} m_x^*, m_y^*, 1, 0, 0, 0, -m_x^* m_x, -m_y^* m_x \\ 0, 0, 0, m_x^*, m_y^*, 1, -m_x^* m_y, -m_y^* m_y \end{bmatrix}$$

$$\mathbf{M_j} h = m_j$$

$$\mathbf{H} = \mathbf{R} + \frac{x}{d^*} n^{*T}$$

- Requires at least **4 coplanar, non-collinear points**(three of them are collinear is ok)
- Forms a system of linear equations from point correspondences
- Uses Singular Value Decomposition (SVD) to estimate h_{33}.
- Estimates and decomposes H to find rotation, translation/depth ratio, and plane normal
- Handles multiple mathematically valid solutions via additional constraints

Essential matrix and eight points

- **Epipolar Constraint:**

$$m_j^T \mathbf{E} m_j^* = 0$$

$$\mathbf{M}e = 0$$

$$\mathbf{E} = [x]_x \mathbf{R}$$

for all matched points

- **8-Point Algorithm Process:**
  - Requires at least **8 non-coplanar** matched points, **no 4 points are coplanar**
  - Forms a system of linear equations from point correspondences
  - Uses Singular Value Decomposition (SVD) to estimate E
  - Decomposes E to recover rotation (R) and translation (x)
  - Validates solutions using positive depth constraint
  - Recovers relative depths of feature points

Thank you